

Table of Contents

[BBACSrcDefTopic](#)

[bbacsrc_about.1.2](#)

[bbacsrc_about.1.3](#)

[bbacsrc_about.1.4](#)

[bbacsrc_about.1.5](#)

[bbacsrc_apps.5.01](#)

[bbacsrc_apps.5.02](#)

[bbacsrc_apps.5.03](#)

[bbacsrc_apps.5.04](#)

[bbacsrc_apps.5.05](#)

[bbacsrc_apps.5.06](#)

[bbacsrc_apps.5.07](#)

[bbacsrc_apps.5.08](#)

[bbacsrc_apps.5.09](#)

[bbacsrc_apps.5.10](#)

[bbacsrc_apps.5.11](#)

[bbacsrc_apps.5.12](#)

[bbacsrc_apps.5.13](#)

[bbacsrc_apps.5.14](#)

[bbacsrc_apps.5.15](#)

[bbacsrc_apps.5.16](#)

[bbacsrc_displays.4.01](#)

[bbacsrc_displays.4.02](#)

[bbacsrc_displays.4.03](#)

[bbacsrc_displays.4.04](#)

[bbacsrc_displays.4.05](#)

[bbacsrc_displays.4.06](#)

[bbacsrc_displays.4.07](#)

[bbacsrc_displays.4.08](#)

[bbacsrc_displays.4.09](#)

[bbacsrc_displays.4.10](#)

[bbacsrc_displays.4.11](#)

[bbacsrc_displays.4.12](#)

[bbacsrc_displays.4.13](#)

[bbacsrc_displays.4.14](#)

[bbacsrc_displays.4.15](#)

[bbacsrc_displays.4.16](#)

[bbacsrc_program.3.01](#)

[bbacsrc_program.3.02](#)

[bbacsrc_program.3.03](#)

[bbacsrc_program.3.04](#)

[bbacsrc_program.3.05](#)

[bbacsrc_program.3.06](#)

[bbacsrc_program.3.07](#)

[bbacsrc_program.3.08](#)

[bbacsrc_program.3.09](#)

[bbacsrc_program.3.10](#)

[bbacsrc_program.3.11](#)

[bbacsrc_program.3.12](#)

[bbacsrc_program.3.13](#)

[bbacsrc_program.3.14](#)

[bbacsrc_program.3.15](#)

[bbacsrc_program.3.16](#)

[bbacsrc_program.3.17](#)

[bbacsrc_too.2.01](#)

[bbacsrc_too.2.02](#)

[bbacsrc_too.2.03](#)

[bbacsrc_too.2.04](#)

[bbacsrc_too.2.05](#)

[bbacsrc_too.2.06](#)

[bbacsrc_too.2.07](#)

[bbacsrc_too.2.08](#)
[bbacsrc_too.2.09](#)
[bbacsrc_too.2.10](#)
[bbacsrc_too.2.11](#)
[bbacsrc_too.2.12](#)
[bbacsrc_too.2.13](#)
[bbacsrc_too.2.14](#)
[bbacsrc_too.2.15](#)
[bbacsrc_too.2.16](#)
[bbacsrc_too.2.17](#)
[bbacsrc_too.2.18](#)
[bbacsrc_too.2.19](#)

BBACSrcDefTopic

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About the BBAC Source

About the BBAC Source

The Broadband Analog Channel Source (BBAC Source) provides AC waveform generation for applications such as audio and communications device testing. Typical tests that can be performed include gain accuracy, frequency response, distortion, and signal-to-noise ratio. The following topics are included:

- Features
- Safety Information

- [Key Terms](#)
- [BBAC Source Software Licensing Requirements](#)

BBAC Source information includes the following sections:

- [BBAC Source Theory of Operation](#)
- [BBAC Source Programming](#)
- [BBAC Source Examples](#)
- [BBAC Source Debug Display](#)

Related Topics

- [BBAC15 Specifications](#)
- [BBACSource](#) (VBT syntax)
- [BBAC DIB Design Guidelines](#)

Two BBAC Source instruments share the BBAC instrument board with a pair of BBAC Capture instruments, which capture AC waveforms. For more information on BBAC Capture, see [About the BBAC Capture](#).

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About the BBAC Source : Features

Features

The BBAC Source instrument can generate high-accuracy, low-noise, and low-distortion signals with frequency components of up to 15MHz. It has a maximum user-programmable sampling rate of 1Gsps and can drive a maximum output amplitude of 5.12Vpk differential.

The BBAC Source offers:

- | | |
|---|---|
| • | 100Ω differential output impedance |
| • | High signal-to-noise ratio and amplitude accuracy |
| • | Programmable DC baseline using Common Mode Voltage |
| • | Mixed-signal pattern microcode control |
| • | Access to Parametric Measurement Unit (PPMU) per pin |
| • | Parameter sets (PSets) for pattern control of instrument settings |
| • | DIB access connectivity |

For BBAC Source hardware specifications, see BBAC15 Specifications.

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About the BBAC Source : Safety Information

Safety Information

When operating the BBAC Source instrument, always follow general UltraFLEX test system safety guidelines. See [Test System Safety](#).

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About the BBAC Source : Key Terms

Key Terms

BBAC	Broadband Analog Channel
DGS	Device Ground Sense
Gsps	Gigasamples per second
HSD	High Speed Digital
ISL	Instrument Sync-Link
ksps	Kilosamples per second
MspS	Megasamples per second
PPMU	Pin Parametric Measurement Unit
PSet	Parameter set
SMEM	Source Memory

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About the BBAC Source : BBAC Source Software Licensing Requirements

BBAC Source Software Licensing Requirements

BBAC instrument hardware can support multiple frequencies across source and capture instruments. However, IG-XL enforces a software licensing scheme that constrains instrument functionality to a specified frequency limit across source instruments and a specified frequency limit across all capture instruments.

When purchasing a software license, you must specify a source bandwidth of 3MHz or 15MHz, a capture bandwidth of 3MHz or 15MHz, and the number of channels in increments of two beyond the first channels (0, 1, 2, 4, 6, 8... 48). The bandwidth value specified applies to all channels of the same type.

For example:

- Source bandwidth: 15MHz
of channels: 8
- Capture bandwidth: 3MHz
of channels: 4

 Note:

The maximum number of source or capture channels available (each) is 2 x (# of available test head slots), or 48 - (2 x # of occupied test head slots).

IG-XL checks for a license on validation of a test program to verify that the number of channels used in the test does not exceed the number licensed. A validation error occurs if more instruments are used in a test than are specified in the license.

IG-XL also verifies the license on execution of a test to verify that the test is not programmed to source or capture signals whose frequency exceeds the licensed limitation. A run-time error occurs

if a test sources or captures at frequencies higher than specified by the license.

A BBAC Source 3MHz license limits the source sample rate to a maximum value of 60Msps. This limits the maximum frequency that can be sourced to 3MHz, due to the requirement that the source sample rate must be 20 times greater than the highest frequency of interest. See [BBAC15 Specifications](#) for the minimum sample rate specifications.

A BBAC Capture 3MHz license limits the capture sample rate to a maximum value of 8Msps. This limits the maximum frequency that can be captured to 3MHz, due to the requirement that the capture sample rate must be 8/3 times greater than the highest frequency of interest. See [BBAC15 Specifications](#) for the minimum sample rate specifications.

License limitations are not enforced when a checker program or calibration routine is running or when a program is running on a simulator.

✓ Note:

BBAC software licenses are node-locked licenses, not floating licenses, that apply to instruments in a specific test system.

Summary of Software Licensing

- All source channels must be the same bandwidth option.
- All capture channels must be the same bandwidth option.
- Source bandwidth is independent of capture bandwidth.
- The number of source channels is limited to increments of two beyond the first instrument (0, 1, 2, 4, 6, 8... 48).
- The number of capture channels is limited to increments of two beyond the first instrument (0, 1, 2, 4, 6, 8... 48).
- Source and capture enabling is not limited. You can order all capture channels and no sources or all source channels and no captures.

For more information, see [IG-XL Licensing](#) in the IG-XL Administration section.



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BBAC Source Examples

BBAC Source Examples

This section contains program examples for the BBAC Source instrument. These IG-XL workbook examples show all IG-XL data and VBT code necessary for a complete running program. The following topics are available:

- [BBAC Source/Capture Loopback](#)
- [BBAC Source Signals VBT Examples](#)
- [BBAC Source PSets VBT Examples](#)

For more information, see:

- [BBACSource](#) (VBT Syntax Reference)
- [BBAC Source Programming](#)

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[BBAC Source Examples](#) : BBAC Source/Capture Loopback

[BBAC Source/Capture Loopback](#)

This workbook example includes the following program components:

- | | |
|---|---------------------------------|
| • | DataTool Sheets |
| • | VBT Code |

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BBAC Source Examples : [BBAC Source/Capture Loopback](#) : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- [Pin Map Sheet](#)
- [Channel Map Sheet](#)
- [Test Instances Sheet](#)
- [Flow Table Sheet](#)
- [Time Sets \(Basic\)](#)

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	Src1_P	Analog
	Src1_N	Analog
	Cap1_P	Analog
	Cap1_N	Analog
	DigControl	I/O

Channel Map Sheet

Channel Map

DIB ID:

View Mode:

Device Under Test		Tester Channel	
Pin Name	Package Pin	Type	Site 0
Src1_P		BBACSrcPos	4.srcpos1
Src1_N		BBACSrcNeg	4.srcneg1
Cap1_P		BBACCapPos	4.cappos1
Cap1_N		BBACCapNeg	4.capneg1
DigControl		I/O	1.ch0

Test Instances Sheet

Test Instances



Test Name	Test Procedure			DC Specs		AC Specs		Time Sets
	Type	Name	Called As	Category	Selector	Category	Selector	
Source_Capture_LoopBack1	VBT	Source_Capture	LoopBack					TSB

Flow Table Sheet

Flow Table



					Flow Domain:	
Label	Enable	Job	Gate		Command	
			Part	Env	Opcode	Parameter
					Test	Source_Capture_LoopBack1

Time Sets (Basic)

Time Sets (Basic)



Timing Mode:

Master Timeset Name:

Time Domain:

Strobe Ref Setup Name:

	Cycle	Pin/Group			Data		Drive				Compare				Edge Resolution
Time Set	Period	Name	 Clock Period	Setup	Src	Fmt	On	Data	Return	Off	Mode	Open	Close	Ref Offset	Mode
ts0	200.0E-09	DigControl		i/o	PAT	NR		0			Off				Auto

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[BBAC Source Examples](#) : [BBAC Source/Capture Loopback](#) : VBT Code

VBT Code

Public Function Source_Capture_LoopBack()

```
Dim SourceSine As New DSPWave
```

```
Dim CaptureWave As New DSPWave
```

```
Dim i As Long
```

```
' Connect BBAC Source to Capture Loopback
```

```
TheHdw.BBACSource.Pins("Src1_P").Connect tIBBACSourceFromPOS Or _  
tIBBACSourceFromNEG, tIBBACSourceToLoopBack
```

```
TheHdw.BBACSource.Pins("Src1_P").Connect tIBBACSourceFromREF, _  
tIBBACSourceToDGS
```

```
TheHdw.BBACCapture.Pins\("Cap1\_P"\).Connect tIBBACCaptureFromPOS Or \_  
tIBBACCaptureFromNEG, tIBBACCaptureToLoopBack
```

```
' Create Sinewave Samples to Source
```

```
Call SourceSine.CreateConstant(0, 65536)
```

```
For i = 0 To 65536 - 1
```

```
SourceSine.Element(i) = Sin((3 * i * 2 * Pi / 65536)) / 2
```

```
Next i
```

```
' Load Setting and Samples into the BBAC and Set up Source Signal
```

```
TheHdw.BBACSource.Pins("Src1_P").Signals.Add "SourceSignal"
```

```
TheHdw.BBACSource.Pins("Src1_P").Signals.DefaultSignal = "SourceSignal"
```

```
TheExec.WaveDefinitions.CreateWaveDefinition "WaveDef", SourceSine, True
```

```
With TheHdw.BBACSource.Pins("Src1_P").Signals("SourceSignal")
```

```
.WaveDefinitionName = "WaveDef"
```

```
.Amplitude = 1
```

```
.Offset = 0
```

```
.SampleSize = 65536
.SampleRate = 655360
.CycleCount = 1
.LoadSettings
End With
' Set up BBAC Capture
TheHdw.BBACCapture.Pins("Cap1_P").Signals.Add "CaptureSignal"
TheHdw.BBACCapture.Pins("Cap1_P").Signals.DefaultSignal = "CaptureSignal"
With TheHdw.BBACCapture.Pins("Cap1_P").Signals("CaptureSignal")
    .SampleSize = 65536
    .SampleRate = 655360
    .CycleCount = 1
    .LoadSettings
End With
' Start Sourcing the signal and capturing with Pattern
' TheHdw.Digital.ApplyLevelsTiming True, True, True
TheHdw.Patterns("./src_cap_loop.pat").Load
TheHdw.Digital.Patgen.TimeoutEnable = False
TheHdw.Patterns("./src_cap_loop.pat").Start "", ""
TheHdw.Digital.Patgen.HaltWait
' Wait for Capture to complete
While TheHdw.BBACCapture.Pins("Cap1_P").IsCaptureInProgress
Wend
' Bind capture data to a DSPwave object
CaptureWave = TheHdw.BBACCapture.Pins("Cap1_P").Signals("CaptureSignal").DSPWave
CaptureWave.Plot "BBAC LOOPBACK"
' Do an SNR Analysis on captured data and display it in immediate window
Debug.Print CaptureWave.Spectrum.CalcSNR
' Disconnect the BBAC Source from the Capture
TheHdw.BBACSource.Pins("Src1_P").Disconnect tlBBACSourceFromPOS Or _
    tlBBACSourceFromNEG, tlBBACSourceToLoopBack
TheHdw.BBACSource.Pins("Src1_P").Disconnect tlBBACSourceFromREF, _
    tlBBACSourceToDGS
TheHdw.BBACCapture.Pins("Cap1_P").Disconnect tlBBACCaptureFromPOS Or _
    tlBBACCaptureFromNEG, tlBBACCaptureToLoopBack
End Function
```

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BBAC Source Examples : BBAC Source Signals VBT Examples

BBAC Source Signals VBT Examples

The following examples use BBAC Source VBT methods and properties:

- [Apply Amplitude and Offset to a BBAC Source Ramp](#)
- [Apply Amplitude and Offset to a BBAC Source Sine Wave](#)
- [Create a Wave Definition for BBAC Source on the Fly](#)
- [Use MST to Set Up, Load, and Start a BBAC Source Signal](#)
- [Use BBAC Source Common Mode Hardware to Introduce an Offset](#)
- [Set All BBAC Source Hardware Signal Parameters to Use Loaded Value](#)
- [Use Direct Programming to Set Up and Start a BBAC Source Signal](#)
- [Use LoadSettings for a BBAC Source Signal and Read the Programmed Settings](#)
- [Use Pattern Control with BBAC Source Default Signal](#)
- [Report Information About a Collection of BBAC Source Signals](#)

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[BBAC Source Examples](#) : [BBAC Source Signals VBT Examples](#) : Apply Amplitude and Offset to a BBAC Source Ramp

Apply Amplitude and Offset to a BBAC Source Ramp

This function applies amplitude and offset to a ramp wave definition.

Public Sub CreateBBACSourceRampSignal()

' This code creates a ramp for a single-ended input

' that goes from 2.5 V to 4.5 V.

' Replace "AIN" with your bbac source (pos) pin name

Dim pinName As String

pinName = "AIN"

' Create a new Signal called "ramp1" on pin 'pinName'.

TheHdw.BBACSource.pins(pinName).Signals.Add "ramp1"

With TheHdw.BBACSource.pins(pinName).Signals("ramp1")

' Specify a Single-Ended device input:

.ConnectionType = tlBBACSourceConnectionTypeSingleEndedNeg

' Program the WaveDefinitionName. Specify "aRamp" on the WaveDef

' sheet

.WaveDefinitionName = "aRamp"

' Program Amplitude to 2.0 - the difference between 4.5 and 2.5

.Amplitude = 2#

' Program Offset to 2.5 - the level to start the ramp at.

' Note: example "Use CommonMode hardware to Introduce Offset"

' shows a way to use hardware to do this, which can improve

' signal quality by using more dynamic range of the converter.

.Offset = 2.5

```
' Program SampleRate and SampleSize; CycleCount defaults to 1.  
.SampleRate = 20000#  
.SampleSize = 50  
' View the samples to verify the Signal is set up.  
.samples.Plot
```

End With

End Sub

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[BBAC Source Examples](#) : [BBAC Source Signals VBT Examples](#) : Apply Amplitude and Offset to a BBAC Source Sine Wave

Apply Amplitude and Offset to a BBAC Source Sine Wave

This function applies amplitude and offset to a sine wave definition for a BBAC Source.

Public Sub CreateBBACSourceUnipolarSineSignal()

' This creates a 1 kHz, 2 Vpk sine wave for a differential input

' that has a DC Offset of 2.5 V.

' Replace "AIN" with your bbac source (pos) pin name

Dim pinName As String

pinName = "AIN"

' Create a new Signal called "sine1" on pin 'pinName'.

TheHdw.BBACSource.pins(pinName).Signals.Add "sine1"

With TheHdw.BBACSource.pins(pinName).Signals("sine1")

' ConnectionType defaults to Differential, so no need to set it.

' Program the WaveDefinitionName. Specify "aSine" on the WaveDef

' sheet

.WaveDefinitionName = "aSine"

' Program Amplitude to 2.0. This is the difference between the

' DC level 2.5 and the peak (0.5 or 4.5).

.Amplitude = 2#

' Program Common Mode Offset to 2.5. This is the DC level.

.CommonMode = 2.5

' Program SampleRate and SampleSize; CycleCount defaults to 1.

.SampleRate = 20000#

.SampleSize = 20

' View the samples to verify the Signal is set up.

.samples.Plot

End With

End Sub

	
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[BBAC Source Examples](#) : [BBAC Source Signals VBT Examples](#) : Create a Wave Definition for BBAC Source on the Fly

Create a Wave Definition for BBAC Source on the Fly

This function uses `CreateWaveDefinition` to make a wave definition from an imported file on the fly.

```
Public Sub ImportFileForBBACSourceSignal()
```

```
    ' Use the FileImport function on the DSPWave to import a pre-existing file
```

```
    ' called "aCustom.txt". Create a wave definition from this file and
```

```
    ' assign it to Signal "customSignal" on pin "AIN".
```

```
    ' Replace "AIN" with your BBAC Source (pos) pin name
```

```
    Dim pinName As String
```

```
    pinName = "AIN"
```

```
    Dim mySamples As New DSPWave
```

```
    TheHdw.BBACSource.pins(pinName).Signals.Add "customSignal"
```

```
    With TheHdw.BBACSource.pins(pinName).Signals("customSignal")
```

```
        mySamples.FileImport "aCustom.txt", File_txt
```

```
        ' Use CreateWaveDefinition to define a wave definition from the
```

```
        ' imported samples on the fly.
```

```
        TheExec.WaveDefinitions.CreateWaveDefinition "myCustom", mySamples
```

```
        ' Set the signals wave definition to the wave definition just created
```

```
        .WaveDefinitionName = "myCustom"
```

```
        ' Set the SampleRate (SampleSize is inherent in the imported data)
```

```
        .SampleRate = 20000
```

```
        ' Amplitude and Offset are built into the wave definition, so use the
```

```
        ' defaults of 1.0 and 0.0 to apply no gain and offset.
```

```
        ' View the samples to verify the Signal is set up.
```

```
        .samples.Plot
```

End With
End Sub

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[BBAC Source Examples](#) : [BBAC Source Signals VBT Examples](#) : Use MST to Set Up, Load, and Start a BBAC Source Signal

[Use MST to Set Up, Load, and Start a BBAC Source Signal](#)

This function uses a [Mixed Signal Timing \(MST\)](#) set to set up, load, and start a 2Vpk differential source signal.

Public Sub UseMSTToSetupAndStartBBACSourceSignal()

' Use MST to set up, load, and start a 2 Vpk diff source signal.

' Replace "AIN" with your bbac source (pos) pin name

Dim pinName As String

pinName = "AIN"

TheHdw.BBACSource.pins(pinName).Signals.Add "sine1"

With TheHdw.BBACSource.pins(pinName).Signals("sine1")

' Specify a Mixed Signal Timing Set for this Test Instance

' before calling ApplyMixedSignalTiming.

.ApplyMixedSignalTiming

' Program Amplitude to 2 V pk.

.Amplitude = 2#

' Use defaults for all other Signal properties

' Load Signal hardware values into hardware

.LoadSettings

' Start sourcing the Signal

.Start

End With

End Sub

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[BBAC Source Examples](#) : [BBAC Source Signals VBT Examples](#) : Use BBAC Source Common Mode Hardware to Introduce an Offset

[Use BBAC Source Common Mode Hardware to Introduce an Offset](#)

This function uses [CommonMode](#) hardware to introduce an offset in a sine wave for a single-ended device input.

Public Sub UseCommonModeForSEBBACSourceSignal()

' Use CommonMode hardware to introduce offset in a sine wave

' for a single-ended device input.

' Replace "AIN" with your BBAC Source (pos) pin name

Dim pinName As String

pinName = "AIN"

TheHdw.BBACSource.pins(pinName).Signals.Add "sine1"

With TheHdw.BBACSource.pins(pinName).Signals("sine1")

' This creates a sine wave for a single-ended device input

' that goes from 0.5 V to 4.5 V.

.ConnectionType = tlBBACSourceConnectionTypeSingleEndedNeg

.WaveDefinitionName = "aSine"

.Amplitude = 2#

.CommonMode = 2.5

' Program SampleRate and SampleSize; CycleCount defaults to 1.

.SampleRate = 20000

.SampleSize = 20

' Load hardware Settings. This will load the CommonMode hardware.

.LoadSettings

' Start sourcing

.Start

End With

' Read Back CommonMode hardware. Set it to 2.5.

Debug.Print TheHdw.BBACSource.pins(pinName).CommonModeVoltage.Value

End Sub

	
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BBAC Source Examples : [BBAC Source Signals VBT Examples](#) : Set All BBAC Source Hardware Signal Parameters to Use Loaded Value

[Set All BBAC Source Hardware Signal Parameters to Use Loaded Value](#)
This function explicitly sets all hardware signal parameters to use the loaded value.

```
Public Sub SetAllBBACSourceSignalHardwareToUseLoaded()  
    ' Instruct all instrument parameters to "Use Loaded Value"  
    ' Replace "AIN" with your BBAC Source (pos) pin name  
    Dim pinName As String  
    pinName = "AIN"  
    TheHdw.BBACSource.pins(pinName).Signals.Add "sine1"  
    With TheHdw.BBACSource.pins(pinName).Signals("sine1")  
        .SampleRate.Mode = tlSignalModeUseLoadedValue  
        .VoltageRange.Mode = tlCSignalModeUseLoadedValue  
        .scalefactormode = tlBBACSourceSignalScaleFactorModeUseLoadedValue  
        .CommonMode.Mode = tlSignalModeUseLoadedValue  
    End With  
End Sub
```

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[BBAC Source Examples](#) : [BBAC Source Signals VBT Examples](#) : Use Direct Programming to Set Up and Start a BBAC Source Signal

Use Direct Programming to Set Up and Start a BBAC Source Signal

This function uses direct programming (as opposed to using loaded values) to set up and start a signal.

Public Sub CreateAndStartBBACSourceSignalWithDirectProgramming()

' Use direct timing programming to set up, load, and start

' a 1 kHz, 2 Vpk diff source signal.

' Replace "AIN" with your BBAC Source (pos) pin name

Dim pinName As String

pinName = "AIN"

TheHdw.BBACSource.pins(pinName).Signals.Add "sine1"

With TheHdw.BBACSource.pins(pinName).Signals("sine1")

.SampleRate = 20000

.SampleSize = 20

.CycleCount = 1 ' 1 is the default

.WaveDefinitionName = "aSine"

.Amplitude = 2

' Use defaults for all other Signal properties

' Load hardware settings for the Signal

.LoadSettings

' Start the Signal

.Start

End With

End Sub

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[BBAC Source Examples](#) : [BBAC Source Signals VBT Examples](#) : Use LoadSettings for a BBAC Source Signal and Read the Programmed Settings

[Use LoadSettings for a BBAC Source Signal and Read the Programmed Settings](#)

This function programs a new signal, uses Load Settings, and reads the programmed hardware settings.

Public Sub ProgramBBACSourceSignalAndReadbackHardware()

```
' Program a new signal, Load Settings, and read the programmed hardware
' settings from Pins language.
' Replace "AIN" with your BBAC Source (pos) pin name
```

```
Dim pinName As String
```

```
pinName = "AIN"
```

```
TheHdw.BBACSource.pins(pinName).Signals.Add "sine1"
```

```
With TheHdw.BBACSource.pins(pinName).Signals("sine1")
```

```
    .SampleRate = 20000
```

```
    .SampleSize = 20
```

```
    .CycleCount = 1
```

```
    .WaveDefinitionName = "aSine"
```

```
    .Amplitude = 2
```

```
    .VoltageRange.Mode = tlCSignalModeUseCalculatedValue
```

```
    .CommonMode.Mode = tlSignalModeUseLoadedValue
```

```
    ' Use defaults for everything else.
```

```
    ' Load hardware settings
```

```
    .LoadSettings
```

```
End With
```

```
' Now that settings are loaded, read back hardware values
```

```
' from Pins language
```

```
With TheHdw.BBACSource.pins(pinName)
    Debug.Print "Loaded SampleRate: " & CStr(.SampleRate)
    Debug.Print "Loaded VoltageRange: " & CStr(.VoltageRange)
    Debug.Print "Loaded CommonMode: " & CStr(.CommonModeVoltage)
End With
End Sub
```

bbacsrc_apps.5.14

	
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[BBAC Source Examples](#) : [BBAC Source Signals VBT Examples](#) : Use Pattern Control with BBAC Source Default Signal

Use Pattern Control with BBAC Source Default Signal

This function uses pattern control with the Default Signal.

```
Public Sub UseBBACSourceDefaultSignalAndPatternControl()  
    ' Replace "AIN" with your BBAC Source (pos) pin name  
    Dim pinName As String  
    pinName = "AIN"  
    ' Replace ".\Patterns\myPat.pat" with your pattern name  
    Dim patternName As String  
    patternName = ".\Patterns\myPat.pat"  
    ' Add 5 Signals to the Signals collection on Pin 'pinName'.  
    ' Give each signal an Amplitude from 0.5 to 2.5.  
    Dim i As Long  
    Dim sigCount As Long  
    Dim sigList() As String  
    For i = 1 To 5  
        TheHdw.BBACSource.pins(pinName).Signals.Add "signal" & CStr(i)  
        With TheHdw.BBACSource.pins(pinName).Signals("signal" & CStr(i))  
            ' Specify a Mixed Signal Timing Set for this Test Instance  
            ' before calling ApplyMixedSignalTiming.  
            .ApplyMixedSignalTiming  
            .Amplitude = 0.5 * i  
        End With  
    Next i
```

```
' Set each Signal to be the default signal and burst the pattern.  
For i = 1 To 5  
    ' Load Settings  
    TheHdw.BBACSource.pins(pinName).Signals("signal" & _  
        CStr(i)).LoadSettings  
    ' Set the DefaultSignal  
    TheHdw.BBACSource.pins(pinName).Signals.DefaultSignal = "signal" & _  
        CStr(i)  
    ' This pattern should have an unnamed "Trig" microcode on pin  
    ' 'pinName'. Run "" will start the pattern from the beginning and  
    ' wait for completion.  
    TheHdw.Digital.Patterns.Pat(patternName).Run ""  
Next i  
End Sub
```

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[BBAC Source Examples](#) : [BBAC Source Signals VBT Examples](#) : Report Information About a Collection of BBAC Source Signals

Report Information About a Collection of BBAC Source Signals

This function reports information about a collection of BBACSource signals using Count and List.

Public Sub ReportInfoAboutBBACSourceSignalsColl()

' Replace "AIN" below with your bbac source (pos) pin name

Dim pinName As String

pinName = "AIN"

' First, add 10 Signals to the Signals collection on Pin 'pinName'

Dim i As Long

Dim sigCount As Long

For i = 1 To 10

 TheHdw.BBACSource.pins(pinName).Signals.Add "signal" & CStr(i)

Next i

[sigCount = TheHdw.BBACSource.pins\(pinName\).Signals.Count](#)

' Print each Signal name

Dim sigList() As String

sigList = TheHdw.BBACSource.pins(pinName).Signals.List

Debug.Print "There are " & sigCount & " signals in the collection " & _
 "on pin '" & pinName & "'. They are:"

For i = 0 To sigCount - 1

 Debug.Print vbCrLf & sigList(i)

Next i

End Sub

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BBAC Source Examples : BBAC Source PSets VBT Examples

BBAC Source PSets VBT Examples

The following examples show the use of BBAC Source PSets syntax.

Create a BBAC Source PSet

```
Public Function CreateBBACSrcPSet(SrcPin As String, PSetName As String, _
    Amplitude As Double, VoltageRange As Double, SampleRate As Double) _
    As Long
    ' Add the PSet to the BBAC Source PSets collection
    TheHdw.BBACSource.Pins(SrcPin).PSets.Add PSetName
    ' Set all the parameters of the PSet(Amplitude, SampleRate, VoltageRange)
    TheHdw.BBACSource.Pins(SrcPin).PSets(PSetName).Set Amplitude, _
        SampleRate, _
        VoltageRange
```

End Function

Use Incremental Programming with PSets

The following example incrementally programs BBAC Source amplitude using a PSet.

```
Public Function AdjustBBACSrcPSetAmplitude(SrcPin As String, _
    PSetName As String, Amplitude As Double) As Long
    ' Add the PSet to the BBAC Source PSets collection
    TheHdw.BBACSource.Pins(SrcPin).PSets.Add PSetName
    ' Set the PSet amplitude
    TheHdw.BBACSource.Pins(SrcPin).PSets(PSetName).Amplitude = Amplitude
    ' Set the SampleRate and Voltage Range to use the loaded value
    TheHdw.BBACSource.Pins(SrcPin).PSets(PSetName).SampleRate.Mode = _
        tIPSetNoChange
    TheHdw.BBACSource.Pins(SrcPin).PSets(PSetName).VoltageRange.Mode = _
        tIPSetNoChange
```

End Function

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BBAC Source Debug Display

BBAC Source Debug Display

The Broadband AC Source (BBAC Source) Debug Display is an interactive diagram of the BBAC Source hardware and signal path, which shows the instrument’s settings and connections in the context of a site. You can change the values, options, and settings in the display and run your test program to see the effects of the changes.

The following topics are available:

- Starting the BBAC Source Debug Display
- BBAC Source Debug Display Interface
- Using the BBAC Source Debug Display

This section describes features specific to the BBAC Source Debug Display. For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

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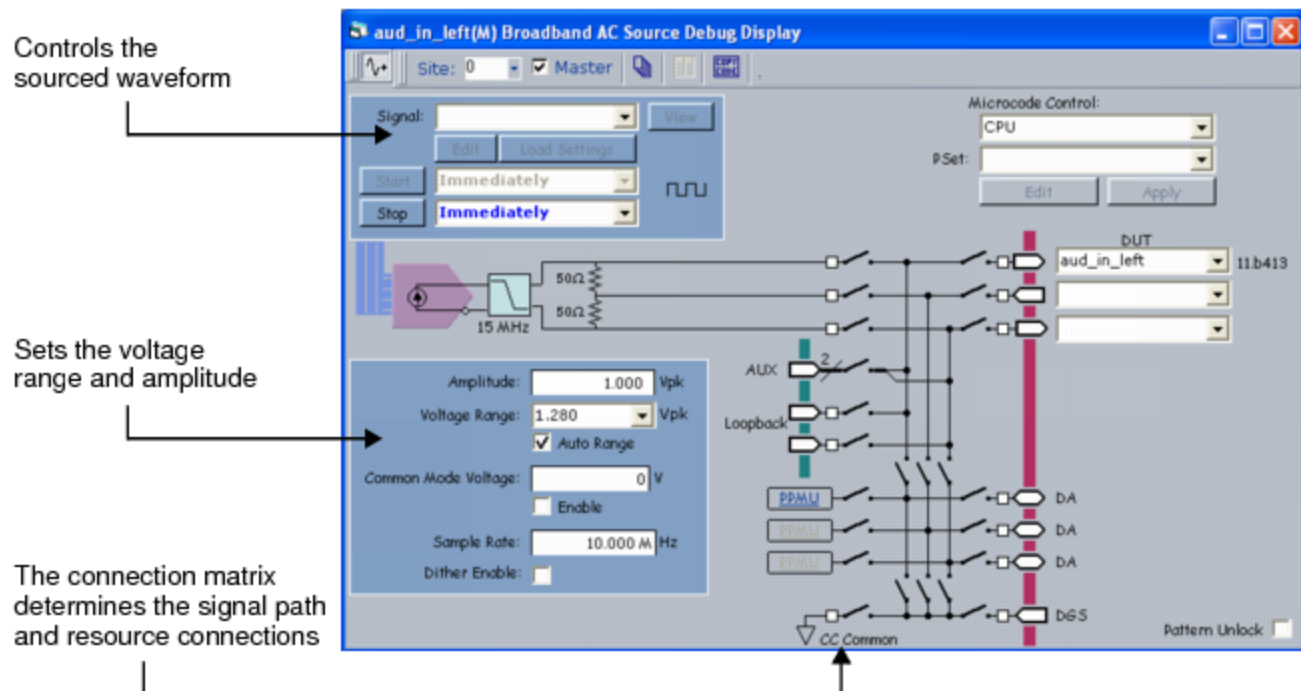
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BBAC Source Debug Display : Starting the BBAC Source Debug Display

Starting the BBAC Source Debug Display

To start the Broadband AC Capture Debug Display:

1. Validate the active test program and ensure that it includes BBAC Source pins.
2. On the IG-XL Tools tab, in the Launch group, select Debug Displays>Broadband AC Source. This opens the display for all active sites.



Broadband AC Source Debug Display

If you start a debug display before the job has been validated, an error appears in the display saying that the program has not been validated. If you open a display for an instrument that is in

the configuration file but not in the channel map, the display shows a message that no instruments have been found.

Once the job is validated, the display shows the current hardware settings. When you run to a breakpoint in the flow or in the VBT code, or as you step through the flow or code, IG-XL updates the display to show any changes to the settings.

For information on other methods for starting debug displays, see [Starting Debug Displays](#) in the Test Debug Environment (TDE) section.

For information on fields used in the BBAC Source debug display, see [BBAC Source Debug Display Interface](#).

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BBAC Source Debug Display : BBAC Source Debug Display Interface

BBAC Source Debug Display Interface

The following sections describe the interface for the BBAC Source debug display:

- | | |
|---|---|
| • | Controlling the Source Signal |
| • | Setting the Voltage Range and Amplitude |

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BBAC Source Debug Display : BBAC Source Debug Display Interface : Controlling the Source Signal

Controlling the Source Signal

The block in the upper-left section of the display includes input boxes and buttons that select which signal to source and where the source instrument will receive the start and stop commands from. You can also use the View button to preview the signal to be sourced in a scope display. Note that most fields and buttons only become active once a signal is specified in the Signal parameter. The table lists BBAC Source Debug Display parameters, buttons, and fields that control sourcing of signals.

BBAC Source Waveform Debug Display Fields

Menu, Button, or Field	Action
Signal	The menu includes the names of the defined signals. If a default signal has been defined, the default signal’s name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT.
View	Brings up the WaveScope tool and plots the sourced signal segment on the WaveScope display. This allows you to visually preview and analyze the signal that is to be sourced. When you are running the program on a test system, WaveScope displays the samples and scales them by the amplitude. When working offline, the WaveScope displays the ideal waveform.

Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.
Load Settings	Loads the signal's timing settings as specified on the Mixed Signal Timing sheet, as well as all of the instrument's settings (for example, voltage range, amplitude, and offset).
Start	Starts sourcing the signal associated with the specified signal according to the action command selected in the menu next to the Start button.
Stop	Stops sourcing according to the action command selected in the menu next to the Stop button. The choices are: • Immediately • At End
	The square wave is a sourcing indicator that moves when the source is sourcing on the tester. This feature only works on line.
Microcode Control	Specifies where the pattern microcode start command, which starts the sourcing of the signal, originates. The menu provides the following options: • CPU: From the host computer • External Trigger: From the DIB • Slaved External Trigger: From another instrument
PSet	Selects a parameter set (PSet). However, it does not apply the setting specified in the PSet to hardware. The PSet is only applied to hardware when you click the Apply button.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor in the TDE section.

Apply	Applies the PSet specified in the PSet field to the sourced signal.
Pattern Unlock	Allows you to use select and monitor pattern pins while that pattern is running (that is, in a CPU loop). See PatternInstrumentUnlock in the VBT Syntax Reference.

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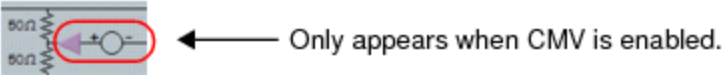
BBAC Source Debug Display : BBAC Source Debug Display Interface : Setting the Voltage Range and Amplitude

Setting the Voltage Range and Amplitude

The block in the lower-left section of the display includes input boxes that allow you to select the voltage and amplitude settings of the output signal. You can also set the common mode voltage to center the AC signal around a common mode voltage, or you can set the Auto Range feature to automatically adjust the voltage range to source the signal within the largest dynamic range. The table lists BBAC Source Debug Display parameters, buttons, and fields that control the voltage and amplitude of signals.

BBAC Source Voltage Range and Amplitude Display Fields

Menu, Button, or Field	Description
Amplitude	Sets the amplitude of the signal as it will be seen by the DUT. Minimum amplitude = 0.0Vpk, differential. Maximum amplitude = 5.12Vpk, differential.
Voltage Range	Sets the voltage range of the signal. The menu provides the available voltage ranges that can be programmed. Minimum voltage range = 0.113Vpk, differential. Maximum voltage range = 5.12Vpk, differential.
Auto Range	When selected, auto range mode is active. As the amplitude changes, the instrument automatically changes the voltage range so that it has the highest signal fidelity by using the entire dynamic range. Using Auto Range results in applying the smallest voltage range that can support the programmed amplitude.

	When auto range is deactivated, the voltage range is left at the last voltage range specified.
Common Mode Voltage	Sets the common mode voltage around which the AC signal is centered. Minimum voltage = -2.5V Maximum voltage = +5.0V
Enable	Enables Common Mode Voltage. When you enable the common mode voltage, you activate the reference buffer, which appears in the debug display. 
Sample Rate	Represents the number of samples to source per second. Minimum sample rate = 5.00kHz Maximum sample rate = 1.00GHz
Dither Enable	Adds dither to the source signal to improve signal distortion (THD and SFDR). See Dither.

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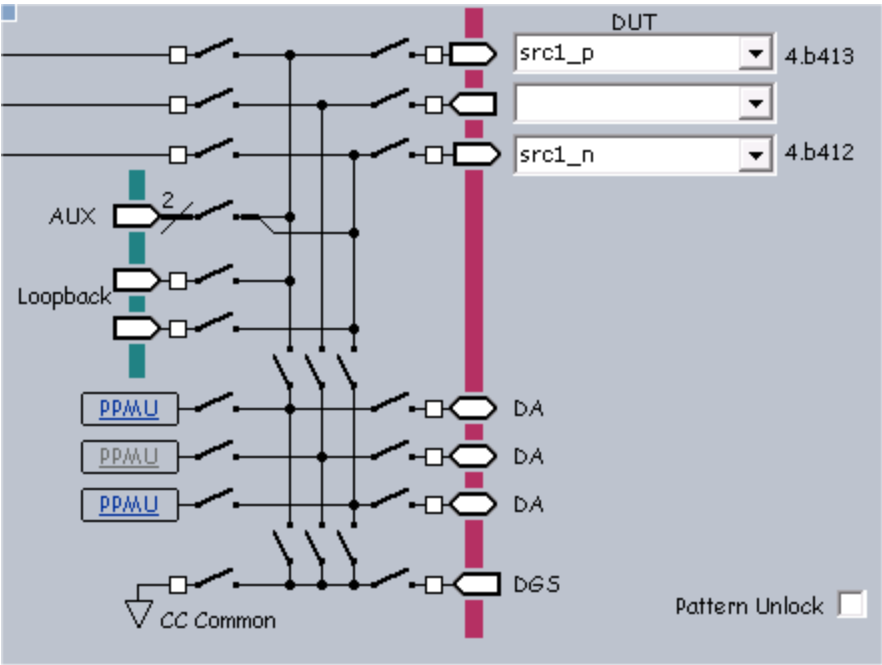
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BBAC Source Debug Display : Using the BBAC Source Debug Display

Using the BBAC Source Debug Display

The figure shows the connections matrix on the Broadband AC Source Debug Display. The connections matrix shows all the possible BBAC Source channel connections to available resources.



BBAC Source Debug Display Connections Matrix

The following topics describe the interface behavior and the actions that you perform to make channel and resource connections in the Broadband AC Debug Display.

- Opening and Closing a Relay Path
- Highlighting a Closed Relay Path

- [Determining a Local or Indirect Channel Connection](#)

The following topics describe the resources that the BBAC Source channels can be connected to:

- [Connecting DUT Pins](#)
- [Connecting DIB Access Channels](#)
- [Connecting Device Ground Sense](#)
- [Connecting the Auxiliary Pin](#)
- [Connecting the Internal Analog Loopback Path](#)
- [Connecting a PPMU](#)
- [Connecting Channel Card Common](#)

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BBAC Source Debug Display : Using the BBAC Source Debug Display : Opening and Closing a Relay Path

Opening and Closing a Relay Path

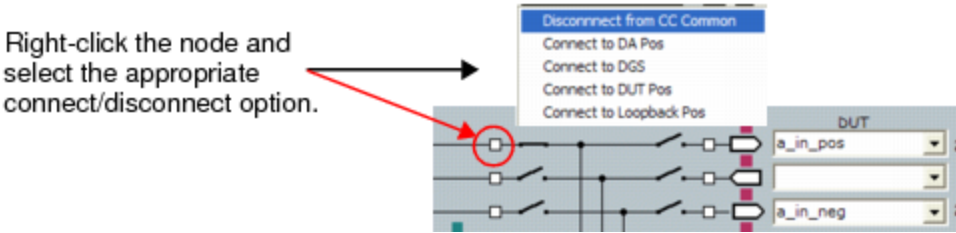
The Broadband AC Source Debug Display simulates channel connections to the DUT and other resources. Graphically, these connections are represented by open and closed relays along the connection path. Clicking a relay does not immediately open or close that relay. It does, however, open a menu that lists the connection options for that relay. You can then select the desired connection action and destination. Note that the relays associated with the Auxiliary and PPMU connections cannot be connected from the BBAC Source Debug Display (see [Connecting the Auxiliary Pin](#) and [Connecting a PPMU](#)).

To open or close a relay path:

1.

Right-click the node along the signal path. A menu appears and lists the connect or disconnect options for that path.
2.

Select the connection destination from the menu. The relay opens or closes to make or break the connection.



Connect/Disconnect Channels and Resources Menu

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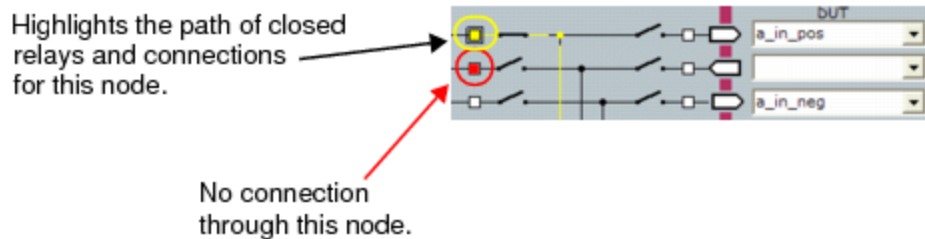
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BBAC Source Debug Display : Using the BBAC Source Debug Display : Highlighting a Closed Relay Path

Highlighting a Closed Relay Path

To determine which resources are connected to a specific BBAC Source channel, you can highlight a connection's relay path.

The figure shows that you can highlight a closed relay path by left-clicking and holding the node along the closed relay path. The path of closed relays and connections that run through that node are then highlighted in yellow. If no connection goes through that node, the node turns red.



Tracing a Closed Relay Path

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BBAC Source Debug Display : Using the BBAC Source Debug Display : Determining a Local or Indirect Channel Connection

Determining a Local or Indirect Channel Connection

The debug display’s connection matrix shows the connection paths related to BBAC Source channels. These connections can result from connections made on this display or those made on another instrument’s debug display. For example, if a BBAC Source pin is connected to a PPMU or the auxiliary connection, the connection is made in the PPMU or primary instrument’s display, and the resulting connection appears in both the BBAC Source and the PPMU/primary instrument’s debug display.

You can determine which BBAC Source channel connections were made locally and which were made by another instrument display. When you click the DUT port of a channel, a menu appears and lists all of the pins behind a particular instrument channel. A check mark next to the pin name indicates that the connection is closed from the BBAC Source instrument to the DUT. An asterisk indicates that the connection was made as a result of another connection.

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BBAC Source Debug Display : Using the BBAC Source Debug Display : Connecting DUT Pins

Connecting DUT Pins

The Broadband AC Source Debug Display allows you to connect an instrument channel and other resources to the DUT pins specified in the test program’s channel map.

Starting the debug display automatically reads in the pins and lists them in the DUT port menus. The DIB channel string is also read into the debug display and is shown beside the appropriate instrument channel.

To connect an instrument channel or resource to a DUT pin:

- | | |
|----|---|
| 1. | Right-click the DUT node. |
| 2. | Select the destination resource or channel from the menu. |

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BBAC Source Debug Display : Using the BBAC Source Debug Display : Connecting DIB Access Channels

Connecting DIB Access Channels

DIB access is an instrument feature that permits a DUT pin to be connected to one instrument board through another instrument board. It allows you to connect multiple resources on different boards to a single pin.

Since DIB access involves connecting instruments, it also means coordinating connections between two instrument debug displays. To distinguish one instrument from another, one is called the “primary” instrument and the other is the “secondary.” The primary instrument is the one that uses another instrument’s DIB access path. The secondary instrument is the one whose path is used by the primary instrument.

In making a DIB access connection, you can only close that connection from the primary instrument’s debug display.

Debug Display DIB Access Connection Example

The following instructions show how to make a DIB access connection between two instrument debug displays. In this case, a BBAC Source is using a BBAC Capture’s DIB access path. That means that the BBAC Source is the primary instrument and the BBAC Capture is the secondary instrument. To make a debug display DIB access connection for this example:

- | | |
|----|--|
| 1. | Start the Broadband AC Source Debug Display. |
| 2. | Select the pin in the DUT Pos menu specified as a DIB access pin. In this case, it is cap1pos. |
| 3. | Right-click the SrcPos node and select Connect to Src Pos from the menu. |

4. [Start the Broadband AC Capture Debug Display.](#)
5. [Select the pin \(cap1pos\) from the DUT Pos menu.](#)

The DIB Access path relays automatically close and the connection is complete.

Since the BBAC Capture is the secondary instrument in this example, you cannot disconnect the DIB access path from within its environment. You must return to the primary instrument, the BBAC Source, to open the DIB access connections.

	
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BBAC Source Debug Display : Using the BBAC Source Debug Display : Connecting Device Ground Sense

Connecting Device Ground Sense

Device ground sense (DGS) is a signal that can be connected to the ground of the DUT. DGS is used as a lowside reference for the PPMUs. Normally, you do not connect source signals to the DGS lines because that will improperly cause current to flow in the sense line.

To connect a DUT pin or instrument channel to the DGS:

- | | |
|----|---|
| 1. | Right-click the DGS node. |
| 2. | Select the destination channel from the menu. |

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BBAC Source Debug Display : Using the BBAC Source Debug Display : Connecting the Auxiliary Pin

Connecting the Auxiliary Pin

The auxiliary connection to the BBAC Source allows another instrument to connect to the BBAC. The auxiliary is not a general purpose instrument interconnection path. It is designed to work with cabling and future product options.

When the auxiliary connection is closed, a BBAC Source is considered a secondary instrument, and the primary instrument is using the BBAC instrument. The auxiliary connections on the debug display are read-only and show whether the auxiliary connections are open or closed.

To open or close the auxiliary connection, make the connection from the primary instrument. Once the auxiliary connection is closed, no other connections on the BBAC Source can be made.

When the auxiliary connection is closed, the Src Pos pin is connected to AUX Pos, Src Ref is connected to CCC, and Src Neg pin is connected to AUX Neg.

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BBAC Source Debug Display : Using the BBAC Source Debug Display : Connecting the Internal Analog Loopback Path

Connecting the Internal Analog Loopback Path

The loopback connection is provided for debugging. It allows you to connect a BBAC Capture directly to a BBAC Source, allowing it to make a measurement of the sourced waveform. The BBAC instrument board has two instrument pairs (BBAC Capture and BBAC Source in Channel 1 and BBAC Capture and BBAC Source in Channel 2). The BBAC Capture Positive in Channel 1 loops back to the BBAC Source Positive of Channel 1. You cannot loop instruments between the two channels (looping BBAC Capture Positive in Channel 1 with BBAC Source Positive of Channel 2). You can independently control the Pos and Neg channels. (For example, you can choose to loop back Positive, Negative, or Both.)

To connect the BBAC Capture to the BBAC Source for loopback purposes:

1.

Right-click the Loopback Positive node and select Connect to Src Pos from the menu.
2.

Right-click the Loopback Negative node and select Connect to Src Neg from the menu.
3.

Right-click the Source Reference node and select CCC from the menu.
4.

Open the Broadband AC Capture debug display.
5.

Right-click the Loopback Positive node and select Connect to Cap Pos from the menu.
6.

Right-click the Loopback Negative node and select Connect to Cap Neg from the menu.

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BBAC Source Debug Display : Using the BBAC Source Debug Display : Connecting a PPMU

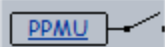
Connecting a PPMU

You can connect each BBAC Source channel to a Pin Parametric Measurement Unit (PPMU) circuit for performing continuity tests.

Although the Broadband AC Source display shows the channel connections to the PPMU circuits, PPMU connections must be made in the PPMU display, not the BBAC Source display.

To connect a BBAC Source DUT pin (Pos, Neg, or Ref) to a PPMU circuit:

1.

 Click the appropriate PPMU link on the Broadband AC Source display based on the BBAC Source channel that is to be connected. This action starts the PPMU debug display.
2.

Select the pin to be connected to the PPMU from the Pin menu.
3.

Click the relay to close the connection.

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BBAC Source Debug Display : Using the BBAC Source Debug Display : Connecting Channel Card Common

Connecting Channel Card Common

You can connect a DUT pin to the channel card common (CCC) ground through the positive, reference, or negative channels.

To connect a DUT pin or instrument channel to the CCC ground:

- | | |
|----|---|
| 1. | Right-click the CC Common node. |
| 2. | Select the destination channel from the menu. |

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BBAC Source Programming

BBAC Source Programming

This section describes programming of a BBAC Source instrument to source data to a DUT. The following topics are available:

- [Simulated Configuration Files](#)
- [Programming Concepts](#)
- [Programming Overview](#)
- [BBAC Source DataTool Programming](#)
- [BBAC Source Pattern Control](#)
- [Using BBAC in Concurrent Testing](#)

For more information, see:

- [BBACSource](#) (VBT Syntax Reference)
- [BBAC Source Examples](#)

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BBAC Source Programming : Simulated Configuration Files

Simulated Configuration Files

You can use the BBAC Source offline by forcing IG-XL to map the BBAC. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- `\tester` folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- `\bin` folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the BBAC board in slot 15:

<Version>

1

</Version>

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#slot[.subslot]	Type	idprom (type serial rev company)
-----------------	------	----------------------------------

15.0	BBAC-15	979-214-00 0000000 0000-A 0000
	BBAC-15	956-375-00 0000000 0000-A 0000

24.0	SupportBoard	974-220-12 0000000 0000-A 0000
	SupportBoard	956-354-00 0000000 0000-A 0000

62.0	DistributionBoard	956-355-00 0000000 0000-A 0000
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To use the [BBAC Source offline](#), the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the BBAC Source online, you need not edit any files. IG-XL automatically maps all boards in the tester. For more information, see [Tester Configuration](#).

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BBAC Source Programming : Programming Concepts

Programming Concepts

Conceptual information about programming methodology and test parameters is provided in the following sections:

- Differential and Single-Ended Measurements
- Dither
- Internal Analog Loopback
- External Trigger
- BBAC Source Reset Values

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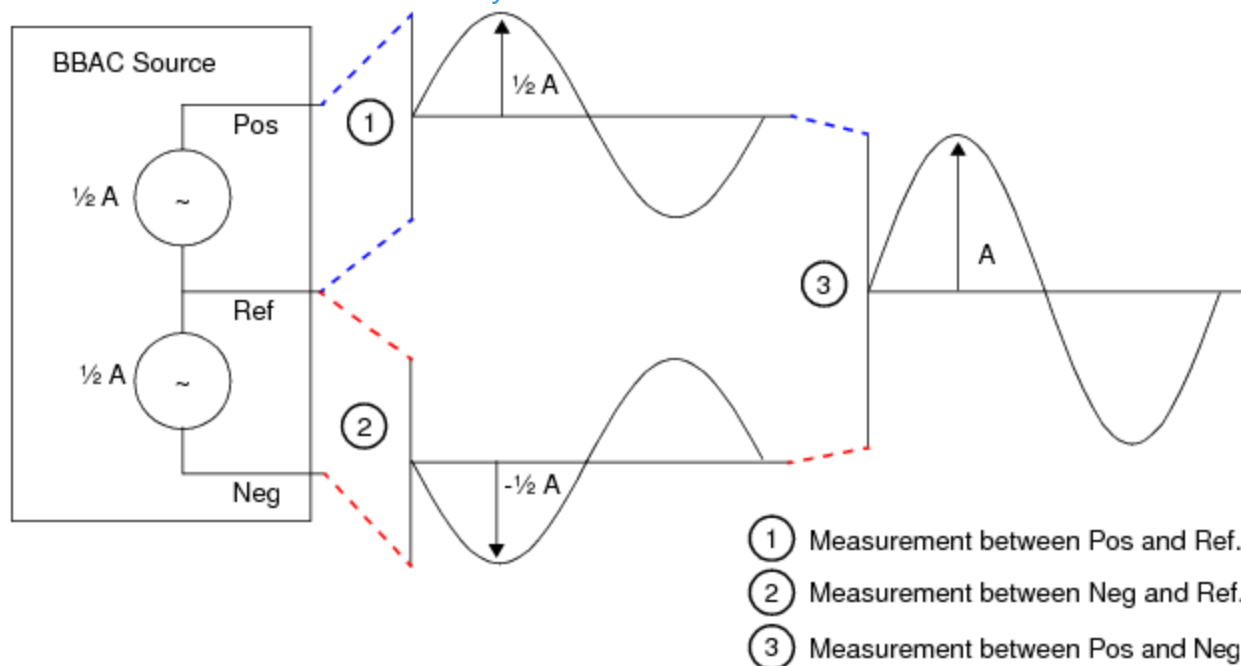
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BBAC Source Programming : Programming Concepts : Differential and Single-Ended Measurements

Differential and Single-Ended Measurements

The figure shows the BBAC Source instrument where "A" is the programmed amplitude. The amplitude represents the differential voltage, measured between the positive (Pos) and negative (Neg) channels. We say, therefore, that the peak voltage (Vpk) is A when measured single-ended and A when measured differentially.



Source Range Measurements

Likewise, the peak-to-peak voltage (Vpp) is defined with respect to the reference as the maximum voltage minus the minimum voltage over time signal. It is A when measured single-ended with respect to the reference and 2A when measured differentially between the positive and negative pins.

Since the BBAC is a differential instrument, BBAC Source ranges and amplitudes are always expressed in differential voltage.

The source range specifies the maximum or minimum voltage that will be generated between the positive and negative channels, which is equivalent to a measurement in Vpk, differential. For example, if the amplitude is set to 1.28V, the range must be set to a value greater than or equal to 1.28V. Setting the range to a value smaller than the amplitude is invalid.

The valid BBAC Source ranges are: 113mV, 160mV, 226mV, 320mV, 452mV, 640mV, 905mV, 1.28V, 1.81V, 2.56V, 3.62V, and 5.12V

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BBAC Source Programming : Programming Concepts : Dither

Dither

A BBAC Source instrument can add dither to the source signal to improve signal distortion (THD and SFDR). Addition of dither to the source signal slightly increases the noise density of the source. Since the dither is noise shaped, the noise density rises more at higher frequencies.

The following topics are available:

- [Enabling and Disabling Dither on a Signal](#)
- [Controlling Dither Hardware Directly](#)

Enabling and Disabling Dither on a Signal

The BBAC Source Signals VBT property Dither Enabled allows you to enable or disable dither on a BBAC Source. Addition of dither to the source signal tends to slightly increase the noise density of the source. Since the dither is noise shaped, the noise density rises more at higher frequencies. The DitherEnabled property defaults and resets to false.

Use the following syntax to enable dither for a signal:

```
TheHdw.BBACSource.Pins().Signals.Item().DitherEnabled.Value = True
```

Use the following syntax to disable dither for a signal:

```
TheHdw.BBACSource.Pins().Signals.Item().DitherEnabled.Value = False
```

☑ **Note:**

Setting DitherEnabled.Value sets DitherEnabled.Mode to tlSignalModeUseValue.

Controlling Dither Hardware Directly

The BBAC Source Dither Enabled property off the Pins node allows you to directly enable or disable the dither hardware. Addition of dither to the source signal tends to slightly increase the noise

density of the source. Since the dither is noise shaped, the noise density rises more at higher frequencies. The DitherEnabled property defaults and resets to false.

Use the following syntax to enable dither:

```
TheHdw.BBACSource.Pins().DitherEnabled = True
```

Use the following syntax to disable dither:

```
TheHdw.BBACSource.Pins().DitherEnabled = False
```

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BBAC Source Programming : Programming Concepts : Internal Analog Loopback

Internal Analog Loopback

The BBAC instrument supports internal analog loopback between the source and capture pairs (Source 1 & Capture 1, Source 2 & Capture 2). The internal analog loopback feature can be used to verify instrument performance and setup conditions without requiring a DIB or external instrumentation.

Each Source and Capture instrument has an analog loopback relay path devoted to this function. You can source and capture a signal within the same instrument board, which eliminates influences outside of the instruments themselves.

The following code demonstrates connecting the BBAC Source to Capture in loopback:

```
Public Function CloseBBACLoopbackRelay(SrcPin As String, _
    CapPin As String) As Long
    TheHdw.BBACCapture.Pins(CapPin).Connect tIBBACCaptureFromInst, _
        tIBBACCaptureToLoopBack
    TheHdw.BBACSource.Pins(SrcPin).Connect tIBBACSourceFromNEG Or _
        tIBBACSourceFromPOS, _
        tIBBACSourceToLoopBack
    TheHdw.BBACSource.Pins(SrcPin).Connect tIBBACSourceFromREF, _
        tIBBACSourceToCCC
```

End Function

The following code demonstrates disconnecting the BBAC Source from Capture in loopback:

```
Public Function OpenBBACLoopbackRelay(SrcPin As String, _
    CapPin As String) As Long
    TheHdw.BBACCapture(CapPin).Disconnect tIBBACCaptureFromInst, _
        tIBBACCaptureToLoopBack
    TheHdw.BBACSource(SrcPin).Disconnect tIBBACSourceFromNEG Or _
        tIBBACSourceFromPOS, _
```

tIBBACSourceToLoopBack

TheHdw.BBACSource(SrcPin).Disconnect tIBBACSourceFromREF, _
tIBBACSourceToCCC

End Function

☒ **Note:**

Ensure that the BBAC source and capture are disconnected from other signal paths (for example, from the DIB/DUT) to avoid external signal interference while using internal loopback.

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BBAC Source Programming : Programming Concepts : External Trigger

External Trigger

The BBAC instrument board includes an external trigger pin for each BBAC instrument, for a total of four pins per board. The external trigger pin is an input pin to the BBAC instrument that allows you to start sourcing a signal or to trigger a capture based on an external event. You can use the external trigger functionality to synchronize the BBAC with other instruments in the tester by sending a signal from the DIB to any of the four AC channels on the BBAC.

Only use an external trigger when you cannot use pattern control for instrument/DUT synchronization. When using an external trigger, the start or trigger is synchronized to an external rising edge of a pulse from a digital channel that is presented to the External Trigger input of a BBAC.

You control the BBAC’s external trigger pins using VBT language statements that allow you to select microcodes (like START, TRIG) to be executed when an event occurs at the trigger input.

BBAC Capture:

- TheHdw.BBACCapture.Pins.ExternalTrigger
- TheHdw.BBACCapture.Pins.MicroCodeControl

BBAC Source:

- TheHdw.BBACSource.Pins.ExternalTrigger
- TheHdw.BBACSource.Pins.MicroCodeControl

 **Note:**

See [BBAC15 Specifications](#) for guidelines and specifications for the minimum time between events and the minimum pulse width when using BBAC external trigger pins.

For more information, see:

- [BBAC Instrument Board](#)
-
- [BBAC Source Pattern Control](#)

	
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BBAC Source Programming : Programming Concepts : BBAC Source Reset Values

BBAC Source Reset Values

The table lists the local reset values for the BBAC Source. Pre-job reset, post-job reset, power-up reset, jobload reset, and user reset all result in the same state, unless otherwise noted.

Reset Values for BBAC Source Settings

Setting	Reset Value
Amplitude	1.0V
Common Mode Voltage	0.0V
Common Mode Enabled	False
Dither	False
Load External Trigger	The load command is removed.
Microcode Control	tlBBACSourceMicroCodeControlByCPU
Sample Rate	10MHz
Voltage Range Autorange	True (Voltage Range = 1.28V)

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BBAC Source Programming : Programming Overview

Programming Overview

To develop a test program that uses the BBAC Source instrument, follow this basic procedure:

1. Create a pin map with pins of the DUT (Programming the Pin Map)
2. Create a channel map that allocates the tester resources to the DUT pins (Programming the Channel Map).
3. Create a mixed signal timing set (Creating a Wave Definition and Mixed Signal Timing Set).
4. Create a test flow, the order in which the test will execute, in the Flow Table sheet.

– Create a test function using VBT code in the Visual Basic editor.

– Instantiate the VBT test and set the test values in the Instance Editor.
5. Validate and run the test program.

For more information, see [BBACSource](#) in the VBT Syntax Reference.

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[BBAC Source Programming](#) : BBAC Source DataTool Programming

[BBAC Source DataTool Programming](#)

This section provides instructions on setting up parameters in worksheets specific to the BBAC Source.

The following topics are included:

- | | |
|---|--|
| • | Programming the Pin Map |
| • | Programming the Channel Map |
| • | Creating a Wave Definition and Mixed Signal Timing Set |

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[BBAC Source Programming](#) : [BBAC Source DataTool Programming](#) : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. Every test program must have a Pin Map sheet. For the BBAC, the connections that you must enter in the pin map are analog channel connections to the DUT.

The recommended Type selection for the BBAC Source instrument is:

Analog	This type is used for the BBAC Source instrument.
--------	---

For more information, see [Pin Map Sheet](#) in the DataTool section.

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BBAC Source Programming : BBAC Source DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For the BBAC Source instrument, you must select one of three channel types from the Type column menu for each pin or pin group. Valid BBAC Source channel types are:

BBACSrcNeg	BBAC Source Negative Pin
BBACSrcPos	BBAC Source Positive Pin
BBACSrcRef	BBAC Source Reference Pin

You must enter the tester channel information value for each site to connect a DUT pin to the instrument. The table lists the available BBAC Source channels. Note that in the table, x indicates the DIB slot number of the BBAC Source.

BBAC Source Channel Resources

BBAC Source Pogo Pin	DIB Channel	Signal Name
BBAC Source Pos 1	x.b413	x.srcpos1
BBAC Source Neg 1	x.b412	x.srcneg1
BBAC Source Ref 1	x.d411	x.scref1
BBAC Source Pos 2	x.f413	x.srcpos2
BBAC Source Neg 2	x.f412	x.srcneg2
BBAC Source Ref 2	x.h411	x.scref2

For more information, see [Channel Map Sheet](#) in the DataTool section.

Information on adapting these instructions for DIB connection options is available in the following sections:

- [Differential Device Input Connections](#)
- [Differential to Single-Ended Conversion Input Connections](#)
- [Single-Ended Device Input Connections](#)
- [Shared Resources](#)
- [DIB Access](#)
- [Multiple Resources](#)

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BBAC Source Programming : BBAC Source DataTool Programming : Creating a Wave Definition and Mixed Signal Timing Set

Creating a Wave Definition and Mixed Signal Timing Set

A BBAC Source instrument generates and delivers analog signals of a specific shape, frequency, and amplitude to the DUT. These signals start as wave definitions, which define the basic shape of the signal. Then a timing solution is added to the waveform definition to create a timing set for the test. The BBAC Source instrument uses the timing set to create a reusable signal to which various amplitudes can be applied.

The following topics provide instructions for creating a wave definition and a timing set:

- | | |
|---|---|
| • | Setting Up the Wave Definitions Sheet |
| • | Creating a Timing Set Using the Mixed Signal Workshop |
| • | Calculating Source Timing |
| • | Source Timing Example: 100Hz Sine Wave |
| • | Source Timing Example: 1.5MHz in High Frequency Mode |
| • | Source Timing Example: 13MHz in High Frequency Mode |

Setting Up the Wave Definitions Sheet

A Wave Definitions sheet contains the wave shape information for waveform definitions that were created using the WaveDesigner. A Wave Definitions sheet stores the wave definition information

and the test program accesses the information when the test is run.

For more information, see [Waveform Definitions](#) and [About WaveDesigner](#) in the WaveDesigner section.

Creating a Timing Set Using the Mixed Signal Workshop

The Mixed Signal Workshop (MSW) tool helps you to build reusable mixed signal timing solutions that meet coherence requirements.

The WaveDesigner tool creates wave definitions, which are independent of timing and instrument specific information. The MSW uses these wave definitions and then applies the instrument specific timing information necessary to generate samples.

The output of the MSW is a mixed signal timing set which contains the signal's structural characteristics and the details of the clocking solution that you created. Timing sets are stored on a Mixed Signal Timing sheet in the IG-XL workbook and become a reusable timing solution for future tests.

For more information, see [Mixed Signal Workshop](#).

Calculating Source Timing

Source timing solutions must adhere to the BBAC Source timing rules.

The minimum F_s allowed is:

$$F_s \geq \text{Maximum [5 kHz or } 20 \times F_i]$$

(1) Where F_i is the highest frequency of interest.

The minimum N allowed is:

$$N \geq \text{Maximum} \left[20 \text{ or } \left(\text{ceiling} \left(F \frac{s}{100 \text{ Msps}} \right) \times 8 \right) \right]$$

(2)

The following figures demonstrate the source timing solution calculations for 100Hz, 1.5MHz, and 13MHz sine waves, respectively.

Source Timing Example: 100Hz Sine Wave

The following example shows the source timing solution calculation for a 100Hz sine wave.

$$F_s \geq \max [5\text{kHz} \parallel 20 \times F_i]$$

$$F_s \geq \max [5\text{kHz} \parallel 20 \times 100\text{Hz}]$$

$$F_s \geq \max [5\text{kHz} \parallel 2\text{kHz}]$$

$$F_s \geq 5\text{kHz}$$

$$N \geq \max [20 \parallel \text{ceiling}(F_s/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel \text{ceiling}(5\text{ksps}/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel 8]$$

$$N \geq 20$$

With $F_s = 5\text{kHz}$ and $M = 1$:

$$N = M \times F_s/F_i$$

$$N = 1 \times 5\text{kHz}/100\text{Hz}$$

$$N = 50 \text{ meets minimum requirement}$$

Therefore, a valid timing solution for a 100Hz sine wave is:

$$F_s = 5\text{kHz}$$

$$N = 50$$

$$M = 1$$

Source Timing Example: 1.5MHz in High Frequency Mode

The following example shows the source timing solution calculation for a 1.5MHz sine wave.

$$F_s \geq \max [5\text{kHz} \parallel 20 \times F_i]$$

$$F_s \geq \max [5\text{kHz} \parallel 20 \times 1.5\text{MHz}]$$

$$F_s \geq \max [5\text{kHz} \parallel 30\text{MHz}]$$

$$F_s \geq 30\text{MHz}$$

$$N \geq \max [20 \parallel \text{ceiling}(F_s/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel \text{ceiling}(30\text{Msps}/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel 8]$$

$$N \geq 20$$

With $F_s = 30\text{MHz}$ and $M = 1$:

$$N = M \times F_s/F_i$$

$$N = 1 \times 30\text{MHz}/1.5\text{MHz}$$

$N = 20$ meets the minimum requirement

Therefore, a valid timing solution for a 1.5MHz sine wave is:

$$F_s = 30\text{MHz}$$

$$N = 20$$

$$M = 1$$

Source Timing Example: 13MHz in High Frequency Mode

The following example shows the source timing solution calculation for a 13MHz sine wave.

$$F_s \geq \max [5\text{kHz} \parallel 20 \times F_i]$$

$$F_s \geq \max [5\text{kHz} \parallel 20 \times 13\text{MHz}]$$

$$F_s \geq \max [5\text{kHz} \parallel 260\text{MHz}]$$

$$F_s \geq 260\text{MHz}$$

$$N \geq \max [20 \parallel \text{ceiling}(F_s/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel \text{ceiling}(260\text{Msps}/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel 24]$$

$$N \geq 24$$

With $F_s = 260\text{MHz}$ and $M = 1$:

$$N = M \times F_s/F_i$$

$$N = 1 \times 260\text{MHz}/13\text{MHz}$$

$N = 20$ does not meet the minimum requirement

So, change M to 2. With $F_s = 260\text{MHz}$ and $M = 2$:

$N = M \times F_s/F_i$

$N = 2 \times 260\text{MHz}/13\text{MHz}$

N = 40 meets the minimum requirement

Therefore, a valid timing solution for a 13MHz sine wave is:

$F_s = 260\text{MHz}$

$N = 40$

$M = 2$

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BBAC Source Programming : BBAC Source Pattern Control

BBAC Source Pattern Control

You can control a BBAC Source instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and parameter sets (PSets).

You create a pattern independently of the test program using the PatternTool or a text editor.

Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling pattern execution.

The following topics are available:

- [BBAC Source Pattern Microcodes](#)
- [BBAC Source ASCII Pattern Syntax](#)

Note the following:

- To use pattern microcodes to control the BBAC Source on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.
- When loading BBAC patterns, do not specify pattern module names in the VBT load command. That is, do not use something like:

```
TheHdw.Patterns("MyPat1.pat:module1").Load
```

Instead, use:

```
TheHdw.Patterns("MyPat1.pat").Load
```

For more information, see:

- [Opening and Managing Pattern Files](#)

- [TheHdw.Patterns.Load](#)

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BBAC Source Programming : BBAC Source Pattern Control : BBAC Source Pattern Microcodes

BBAC Source Pattern Microcodes

A microcode controls a single function, such a starting or stopping the sourcing of a signal. Extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is sourced. To establish repeatable timing, or to synchronize with other instruments (such as HSD) in the tester, use a pattern. If you want to start generating a signal but are not concerned with precise timing, however, then you do not need a pattern. The table lists the extended microcodes used to issue commands to a BBAC Source instrument from a pattern.

BBAC Source Pattern Microcode	
Microcode	Description
PSet <PSetName>	Executes the specified parameter set (PSet) which changes the instrument’s setup (voltage range, amplitude, and sample rate). This microcode requires a single argument, which is the name of the PSet to be applied.
Resync	Forces phase alignment of the BBAC source instrument clock with the HSD T0 clock.
Start <SignalName>	The instrument immediately (on the next available logical analog clock) starts sourcing the named signal in SMEM and repeats it continuously until a Stop microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the

	default signal is used (see TheHdw.BBACSource.Pins.Signals.DefaultSignal).
StartE <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the specified signal in SMEM and repeats it continuously until a Stop microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.BBACSource.Pins.Signals.DefaultSignal).
Start1 <SignalName>	The instrument immediately starts sourcing (on the next available logical analog clock) the specified signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.BBACSource.Pins.Signals.DefaultSignal).
Start1E <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the specified signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.BBACSource.Pins.Signals.DefaultSignal).
Stop	The instrument stops sourcing the currently sourced signal immediately. Stop does not affect a signal that is running once.
StopE	The instrument stops sourcing the currently sourced signal at the end of the signal.

A parameter set (PSet) is a unique microcode that contains a set of values that describe a specific instrument setup. PSets allow a test program to change instrument setups within a pattern execution. Although a PSet is a hardware capability that changes the setup of an instrument during a pattern burst, you define it using VBT language and invoke it using PSet microcode.

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BBAC Source Programming : BBAC Source Pattern Control : BBAC Source ASCII Pattern Syntax

BBAC Source ASCII Pattern Syntax

BBAC Source pattern microcodes are instructions you can place on an individual vector of a pattern. For that vector, a microcode instructs the BBAC Source to perform a particular action, such as starting a new signal.

You can only use BBAC Source microcodes with pins that the instruments statement has specified as BBACSource. For example:

```
instruments = {
    DUT_AIN:BBACSource;
}
```

For more information, see [Instruments Statement](#) in the Pattern Language section.

If a vector contains any BBAC Source related microcode other than nop, that vector is placed in SRM.

The format for specifying a BBAC Source microcode on a particular vector is:

(pin-list : BBACSource = microcode)

The following code shows the ASCII pattern microcode syntax for the BBAC Source and Capture instruments based on the pattern in [BBAC Pattern Microcodes Example](#). The device analog input (DUT_AIN) and analog output (DUT_AOUT) are connected to a BBAC Source and a BBAC Capture respectively.

```
instruments = {
    DUT_AOUT:BBACCapture;
    DUT_AIN:BBACSource;
}
```

```
vector ( $tset, Digital1, Digital2)
{
    > tset1 0      1      ;
```

```

(DUT_AOUT = Resync)
(DUT_AIN = Resync)
> - 1 1 ;
(DUT_AOUT = Trig )
(DUT_AIN = Start )
> - 0 1 ; /* Trigger/Start the default signal */
(DUT_AOUT = Trig Audio_1kHz)
(DUT_AIN = Start Audio_1kHz)
> - 1 1 ; /* Trigger/Start a named signal */
(DUT_AOUT = Enable_Alarm)
(DUT_AIN = Start1 )
> - 0 1 ;
(DUT_AOUT = Disable_Alarm)
(DUT_AIN = StartE )
> - 1 1 ;
(DUT_AOUT = Enable_Inst_Cond)
(DUT_AIN = Start1E )
> - 0 - ;
(DUT_AOUT = Disable_Inst_Cond)
(DUT_AIN = Stop)
> - 1 - ;
(DUT_AIN = StopE)
> - 0 - ;
(DUT_AOUT = PSet AudioPSet)
(DUT_AIN = PSet AudioPSet)
> - 1 - ;
> - - - ;
}

```

Comment - 4

TSet	Command	Label	Vector/Cycl	DUT_AOUT (BBACapture)	DUT_AIN (BBACSource)	Digital1	Digital2	Comment
tset1			0			0	1	
-			1	Resync	Resync	1	1	
-			2	Trig	Start	0	1	Trigger / Start the default signal
-			3	Trig Audio_1kHz	Start Audio_1kHz	1	1	Trigger / Start a named signal
-			4	Enable_Alarm	Start1	0	1	
-			5	Disable_Alarm	StartE	1	1	
-			6	Enable_Inst_Cond	Start1E	0	-	
-			7	Disable_Inst_Cond	Stop	1	-	
-			8		StopE	0	-	
-			9	PSet AudioPSet	PSet AudioPSet	1	-	
-			10			-	-	

Site: 0 (IA) No pattern started Stopped at Vec #: 0; Cycle #: 0 Stopped Not Loaded Pass Vertical: 11 Vectors Vectors 0 to 10

BBAC Pattern Microcodes Example

 Note:

The pattern microcodes in the figure are spaced too closely together to be executed by the Pattern Generator. The minimum spacing required by the pattern generator depends on the BBAC sample rate and the pattern vector rate. For more information, see BBAC15 Specifications.

Related Topics

- | | |
|---|--|
| • | Opening and Managing Pattern Files |
| • | Pattern Compiler |
| • | Pattern Reverse Compiler |

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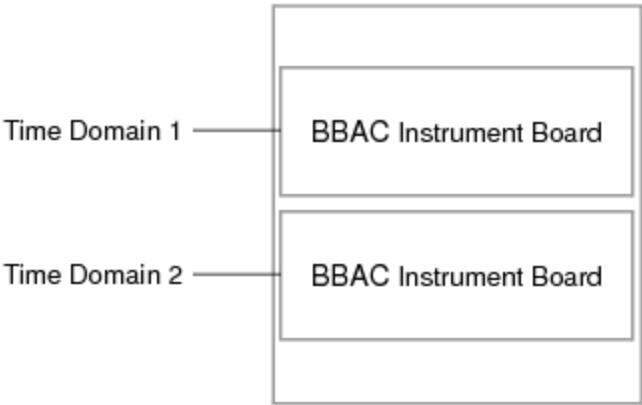
BBAC Source Programming : Using BBAC in Concurrent Testing

Using BBAC in Concurrent Testing

To support concurrent testing, you can assign each BBAC instrument to a unique time domain. Concurrent features for BBAC include:

- Each BBAC board can be configured for one time domain
- Each BBAC instrument can be controlled by a unique time domain using Sync-Link. Features include:
 - Independent access to Sync-Link
 - Independent sample clock programming, one sample rate per channel
 - Independent clock synchronization with other BBAC instruments or with digital clocks
 - Independent control by digital pattern
 - Independent capture memory and triggers
- Multiple instruments can be grouped into a common time domain
- BBAC is restricted to using the Global flow domain and synchronous pattern starts

The following figure shows a configuration with each BBAC instrument assigned to its own time domain.



BBAC Concurrent Testing Configuration Example, 2 Time Domains
For more information, see [About Concurrent Test](#).

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BBAC Source Theory of Operation

BBAC Source Theory of Operation
Conceptual and reference information on the BBAC Source instrument’s hardware and signal path is provided in the following sections:

- | | |
|---|---|
| • | BBAC Instrument Board |
| • | BBAC Source Signal Path |
| • | Connectivity Options |
| • | User View of DIB Pad Patterns |

For information on BBAC Source instrument performance specifications and standards, see [BBAC15 Specifications](#).

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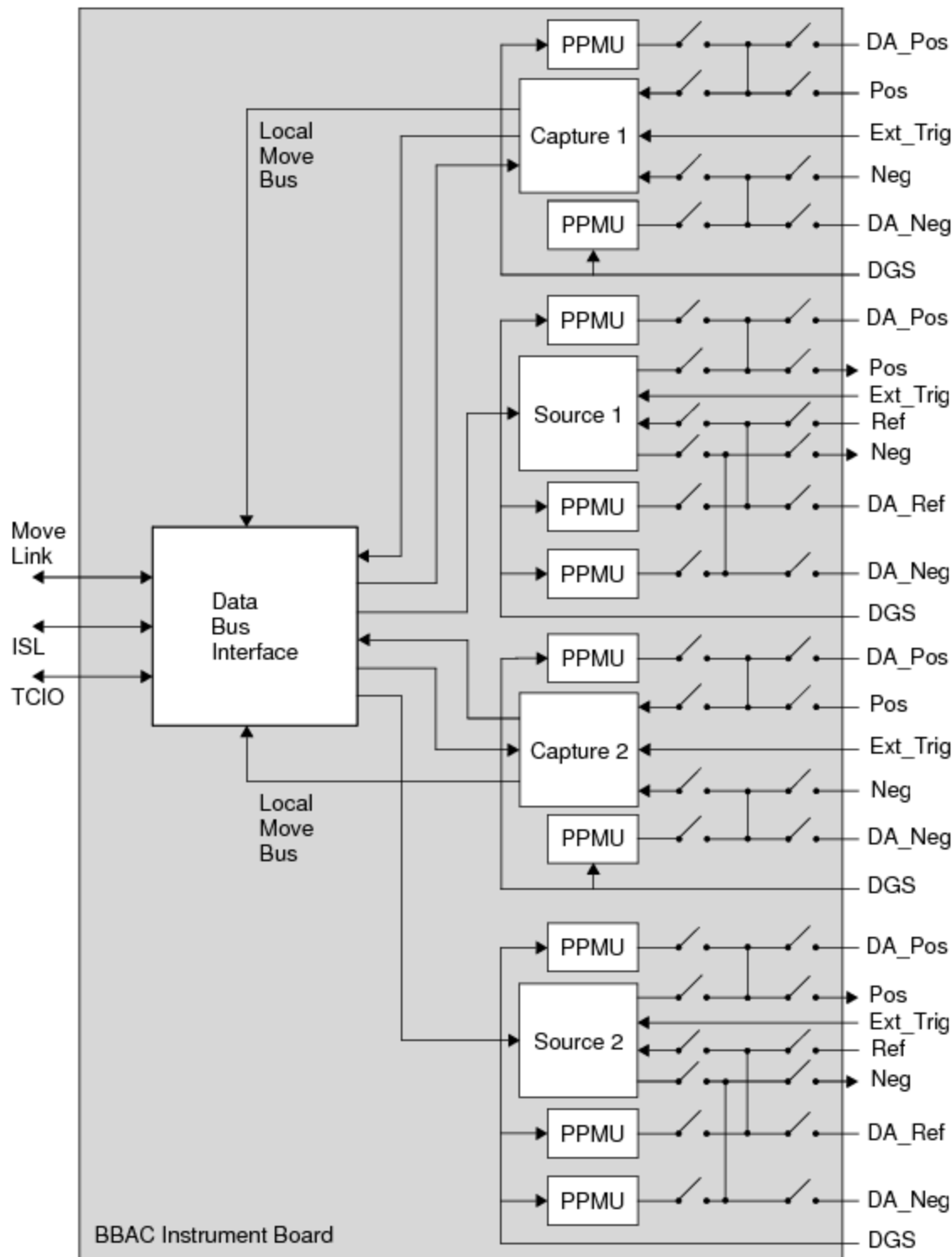
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[BBAC Source Theory of Operation](#) : BBAC Instrument Board

BBAC Instrument Board

The figure shows a block diagram of the BBAC instrument board. The board occupies a single slot in the UltraFLEX test head and contains two BBAC Source instruments and two BBAC Capture instruments.



BBAC Instrument Board

Features of the BBAC instrument board include:

- Ten Pin Parametric Measurement Unit (PPMU) instruments, which can be used for parametric testing
- MoveLink connections from the BBAC Capture instruments to one or more dedicated DSP computers in the test system for high speed transfer and DSP processing of captured data

- Instrument Sync-Link (ISL) connections from each BBAC instrument to the Pattern Generator in the High Speed Digital instrument for accurate and repeatable synchronization
- DIB access connections per instrument channel
- An internal analog loopback connection between each BBAC Source and Capture instrument pair (Source1 and Capture1, Source2 and Capture2)
- External trigger connections for starting a BBAC Source and triggering a BBAC Capture
- One Device Ground Sense (DGS) connection shared by the PPMU circuits for each Source and Capture channel

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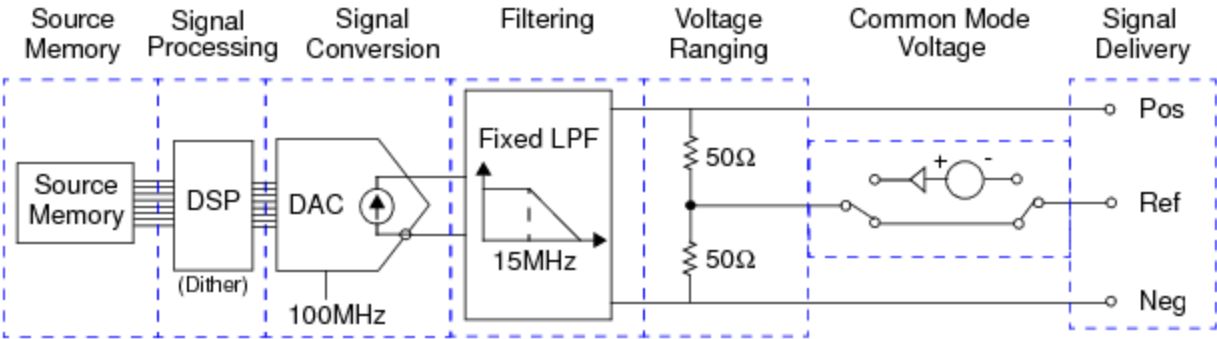
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BBAC Source Theory of Operation : BBAC Source Signal Path

BBAC Source Signal Path

A BBAC Source instrument provides analog signals of a specific shape, frequency, and amplitude to a device under test (DUT). This figure shows how the BBAC Source instrument hardware generates and delivers an analog signal.



BBAC Source Components

This process is described in the following topics:

- Source Memory
- Signal Processing
- Signal Conversion
- Filtering
- Voltage Ranging

- [Common Mode Voltage](#)

- [Signal Delivery](#)

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BBAC Source Theory of Operation : BBAC Source Signal Path : Source Memory

Source Memory

The BBAC Source memory is 4M samples deep. However, the largest single segment length is 2M samples. IG-XL loads source segments as required by host computer and pattern starts. At any one time, up to 500 segments can be loaded in SMEM.

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BBAC Source Theory of Operation : BBAC Source Signal Path : Signal Processing

Signal Processing

The BBAC Source uses a series of digital signal processing stages to prepare the data for conversion by the DACs. First, the data is resampled to convert the sample rate from the user-specified sample rate to the instrument's fixed sample rate. The minimum programmable sample rate is 5ksps, and the maximum sample rate is 1Gsps (see [Calculating Source Timing](#)). To achieve the instrument's performance specifications, however, the following rules determine the source sample rate and the number of samples in a source segment:

The minimum F_s allowed is:

$$(1) \quad F_s \geq \text{Maximum [5 kHz or } 20 \times F_i]$$

Where F_i is the highest frequency of interest.

The minimum N allowed is:

$$(2) \quad N \geq \text{Maximum [20 or ceiling } \left(\frac{F_s}{100 \text{ Msps}} \right) \times 8]$$

See [Calculating Source Timing](#) for examples of how to calculate the Source timing values while respecting the rules.

The Mixed Signal Workshop (MSW) tool provides timing solutions that meet the instrument's sample rate and sample size requirements. For more information, see [Mixed Signal Workshop](#). An amplitude scaling stage then adjusts the amplitude of the data. Calibration and noise-shaping are also applied. Finally, dither can optionally be added.

Adding dither to the source signal improves the signal distortion (THD and SFDR). Addition of dither tends to cause a slight increase in the noise density of the source. Since the dither is noise shaped, the noise density rises more at higher frequencies.

The default and reset state for dither is disabled. For more information, see [Dither](#).

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BBAC Source Theory of Operation : BBAC Source Signal Path : Signal Conversion

Signal Conversion

The BBAC Source instrument’s digital-to-analog converter (DAC) converts digital samples to an analog signal at a fixed sample rate of 100Msps.

Clock Resolution

The BBAC has a period-based clock. Resolution is defined in terms of period resolution and therefore is frequency dependent. The BBAC frequency is governed by the equation:

BBAC Clock Frequency = $\frac{100\text{MHz} \times 2^{40}}{2^{40} \times (I + J)}$

Where: $I \geq 1, 0 < J < 2^{40}$
(3)

The 40 bits of clock resolution allows the both the BBAC Source and BBAC Capture instruments to achieve any desired frequency to a degree of precision that far exceeds the digital subsystem. While the BBAC is not always guaranteed to be coherent with the digital subsystem in the strict sense (a perfect integer ratio relationship), any source or measurement error introduced due to coherence error will be less than can be measured by the BBAC analog performance.

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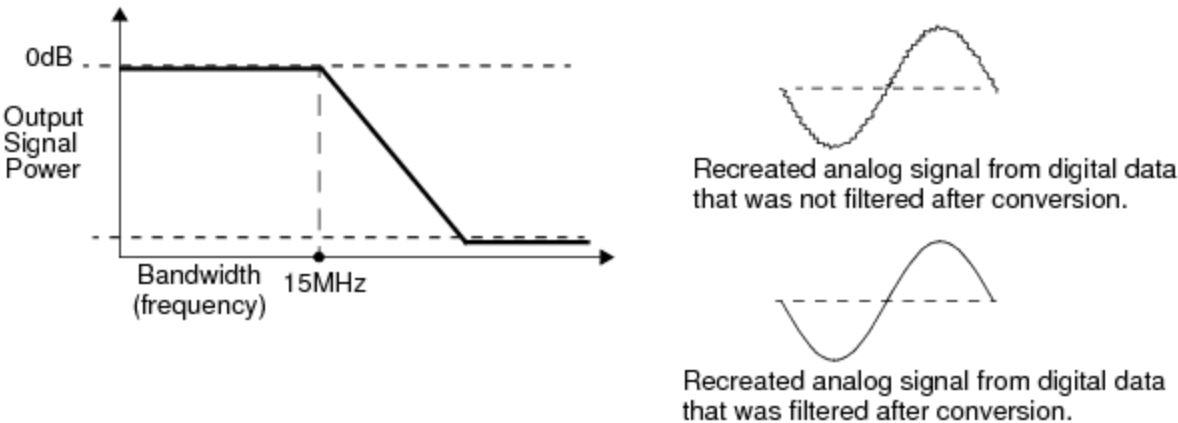
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BBAC Source Theory of Operation : BBAC Source Signal Path : Filtering

Filtering

The source signal passes through a 15MHz low-pass filter, which removes high frequency quantization noise from the synthesized signal and smooths the sequence of discrete analog levels into a continuous analog signal. Since the DAC always operates at 100Msps, only one preset low-pass filter is required with pass band ending at 15MHz. This filter is fixed and cannot be switched or controlled. With the conversion always running at a high speed, the noise shaping process does not generate large amounts of near out-of-band noise, thus producing a high fidelity signal.



15MHz Low-Pass Filter Reduces High Frequency Quantization Noise

For filter specifications, see the BBAC15 Specifications.

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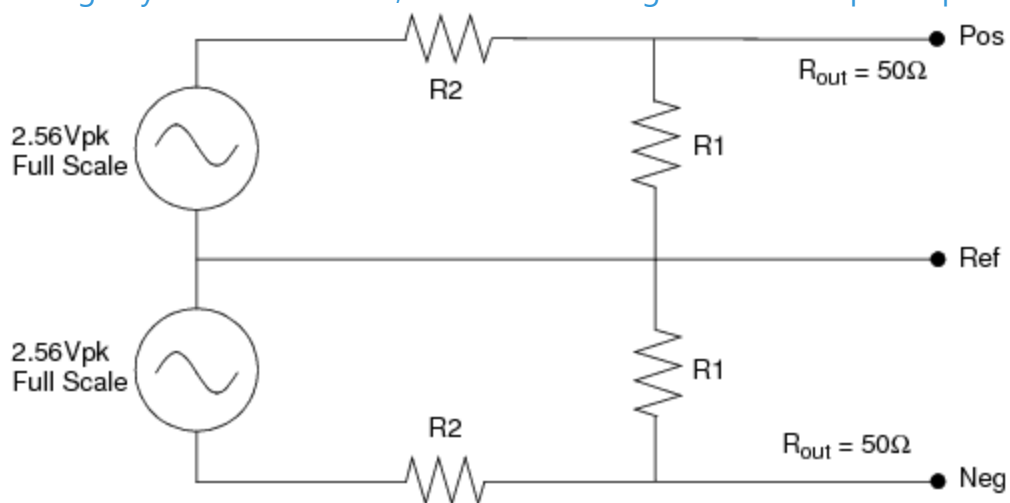
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BBAC Source Theory of Operation : BBAC Source Signal Path : Voltage Ranging

Voltage Ranging

The analog signal is attenuated by the range circuitry to achieve the programmed amplitude. The range circuitry is implemented as a resistor structure, which is designed to attenuate the full-scale voltage by -3dB increments, while maintaining a 50ohm output impedance.



$$R_{out} = R1 \parallel R2 = 50\Omega \text{ relative to Reference}$$

Voltage Ranging Circuit

Since the BBAC Source is nominally 50ohms (100ohms differentially), the amplitude of the signal depends on the input impedance of the DUT. If the impedance is infinite, the DUT will see the full swing for the attenuator setting. Combinations of the AC and DC portion of the signal must also fall within the output compliance of each individual pin. Combinations that exceed the maximum output values are clipped.

For the BBAC Source's output compliance values, see the BBAC15 Specifications.

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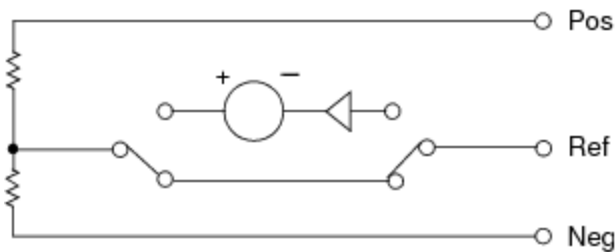
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BBAC Source Theory of Operation : BBAC Source Signal Path : Common Mode Voltage

Common Mode Voltage

Use common mode voltage to bias the source signal around the device's bias point. The BBAC Source reference pin and baseline circuitry provide common mode offset to the source signal. The figure shows the BBAC Source common mode offset circuitry.

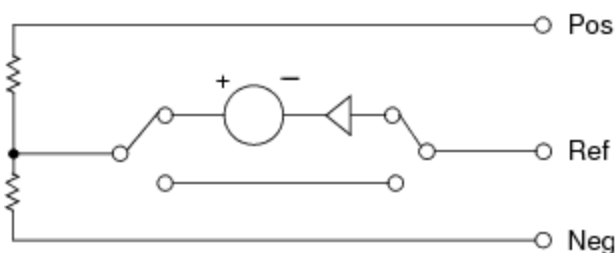


Common Mode Voltage Circuit

This figure also shows that the input reference signal can be used to directly provide a common mode offset to the source signal. In typical applications, the reference pin is connected to either the DIB ground, DUT ground, or another device voltage reference.

Note that the signal path in the figure does not have a buffer at the input of the reference pin and, as such, it is not a high impedance input. Accordingly, if an unbuffered device output is to be connected to the reference pin, either buffer it on the DIB or use the baseline circuitry with a 0.0V offset.

This figure shows that an additional DC offset can be added, in addition to the voltage present at the reference pin, by using the baseline circuitry.



Adding Additional DC Offset

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BBAC Source Theory of Operation : BBAC Source Signal Path : Signal Delivery

Signal Delivery

A BBAC Source delivers the analog signal to the DUT through the source positive and negative pins, bbacsrc_pos and bbacsrc_neg. The reference for the source is the input seen at the reference pin. In addition to the signal pins, the Source instrument’s positive and negative pins have a shield. Typically, you route this shield adjacent to the signal traces to provide a physical shield for the source signal. Best practice for the grounding of these shields is application dependent.

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BBAC Source Theory of Operation : Connectivity Options

Connectivity Options

The following topics describe the connectivity options for connecting a BBAC Source instrument to resources through the DIB:

- [Differential Device Input Connections](#)
- [Differential to Single-Ended Conversion Input Connections](#)
- [Single-Ended Device Input Connections](#)
- [Shared Resources](#)
- [DIB Access](#)
- [Multiple Resources](#)
- [Using RefGround to Connect a DUT Pin to Instrument Ground](#)

Note:

DGS pins are intended as a ground reference for the PPMUs and not for the BBAC Source instrument. Misapplication of a DGS connection can result in the instrument not realizing the instrument's performance specifications.

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BBAC Source Theory of Operation : Connectivity Options : Differential Device Input Connections

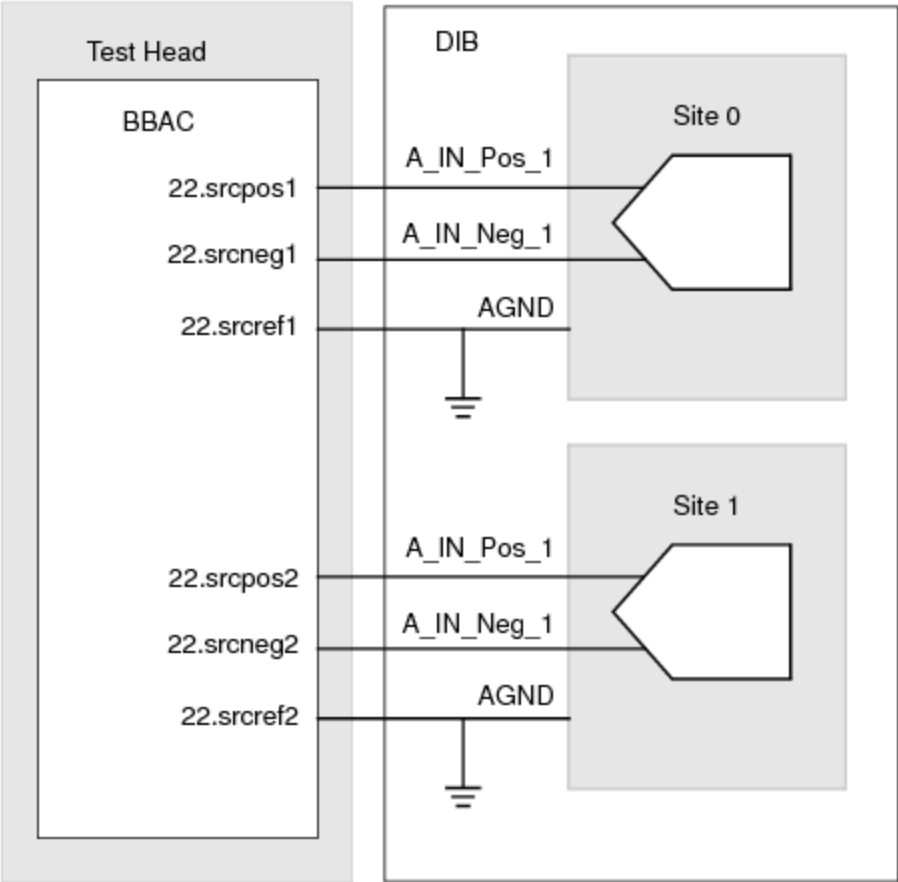
Differential Device Input Connections

The two primary ways to make a differential connection between a BBAC Source instrument and a device with differential inputs are:

- | | |
|---|--|
| • | Source Reference Connected to DIB Ground |
| • | Source Reference Connected to Device Reference Pin |

Source Reference Connected to DIB Ground

The figure shows that the source reference pin is connected to the DIB ground. As such, it does not appear as a reference in either the pin map or the channel map, as shown in the tables.



Source Reference Connected to DIB Ground

Pin Map for Differential Connection: Source Reference to DIB Ground

Group Name	Pin Name	Type
	A_IN_Pos_1	Analog
	A_IN_Neg_1	Analog

Channel Map for Differential Connection: Source Reference to DIB Ground

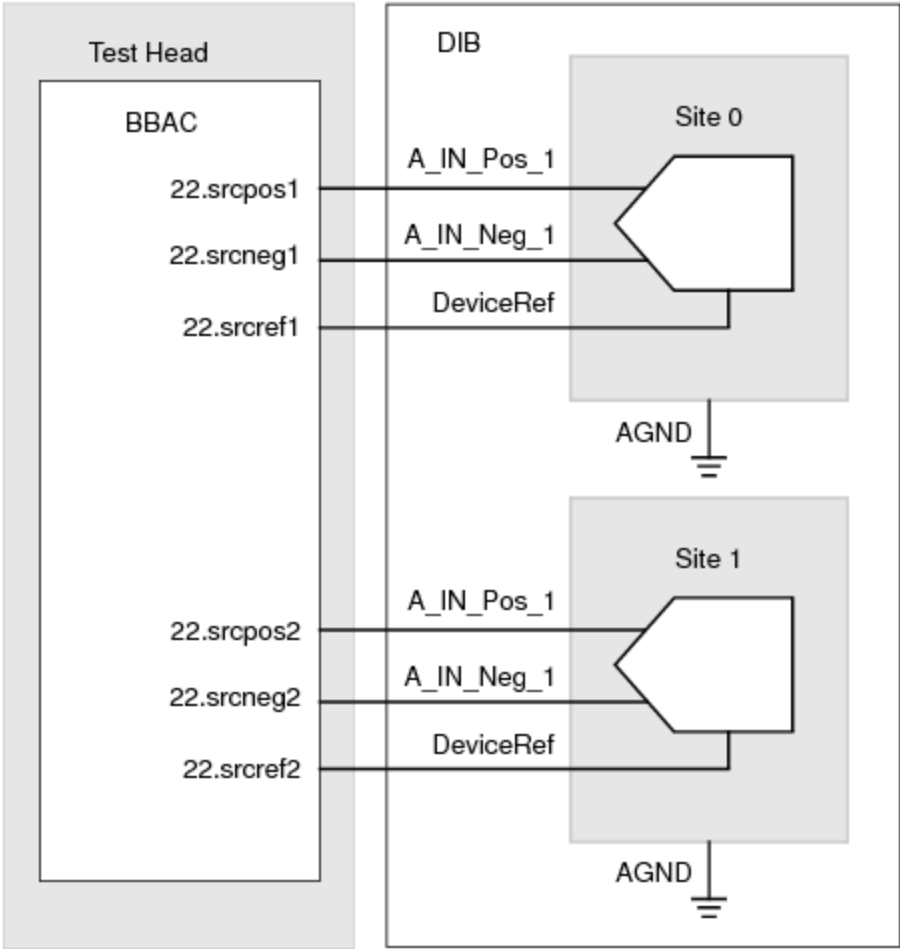
Pin Name	Package Pin	Type	Site 0	Site 1
A_IN_Pos_1	BBACSrcPos		22.srcpos1	22.srcpos2
A_IN_Neg_1	BBACSrcNeg		22.srcneg1	22.srcneg2

Although the reference pin is connected to the DIB ground, you must still close the source reference pin’s connection to the DIB when programming the instrument using VBT statements. Even though the reference pin is not mentioned in the pin map or channel map, you can still do this because the source instrument can be programmed using any pin of that channel. Note that

use of the reference pin’s PPMU and DIB Access features are not supported if it’s not specified in the channel map.

Source Reference Connected to Device Reference Pin

The figure shows the BBAC Source reference channel connected to a device pin that provides a reference voltage for the BBAC Source. As such, this pin appears in the pin map and the channel map.



Source Reference Connected to Device Reference Pin

Pin Map for Differential Connection: Source Reference to Device Reference

Group Name	Pin Name	Type
	A_IN_Pos_1	Analog
	A_IN_Neg_1	Analog
	DeviceRef	Analog

Channel Map for Differential Connection: Source Reference to Device Reference

Pin Name	Package Pin	Type	Site 0	Site 1
A_IN_Pos_1		BBACSrcPos	22.srcpos1	22.srcpos2
A_IN_Neg_1		BBACSrcNeg	22.srcneg1	22.srcneg2
DeviceRef		BBACSrcRef	22.srcref1	22.srcref2

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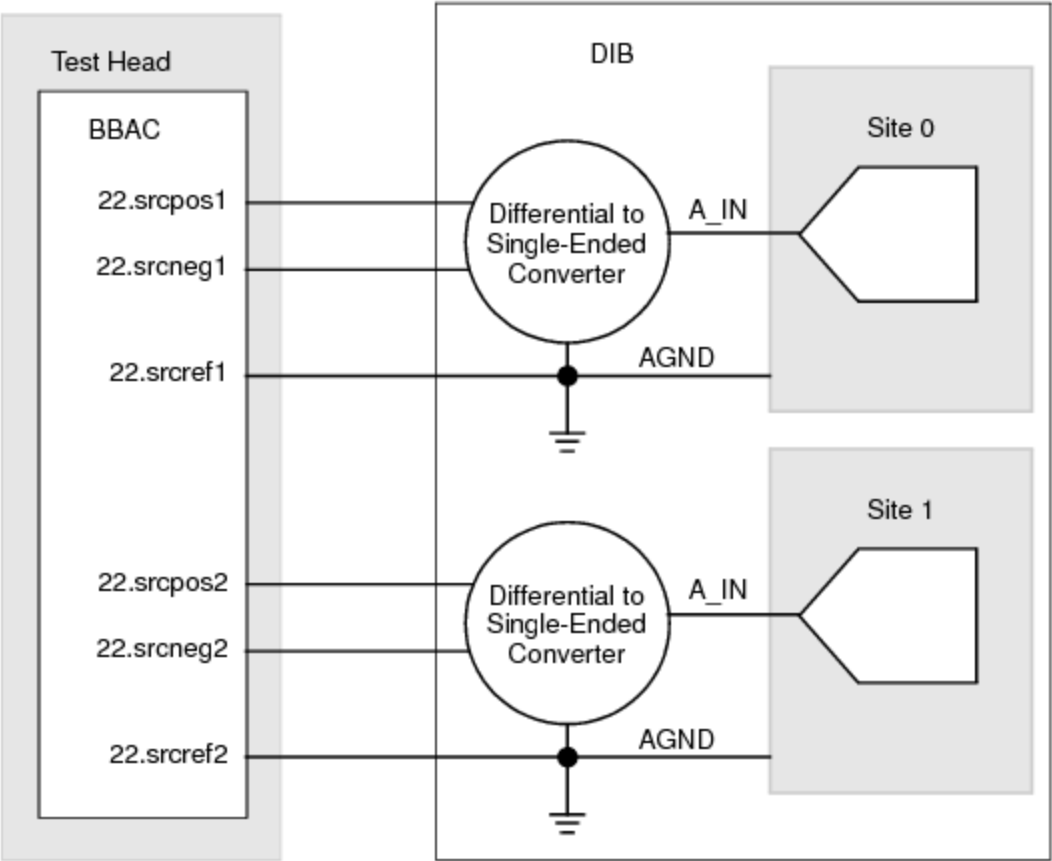
[BBAC Source Theory of Operation : Connectivity Options](#) : Differential to Single-Ended Conversion Input Connections

[Differential to Single-Ended Conversion Input Connections](#)

Use the [Single-Ended DUT Pin Only](#) connection scheme to connect a BBAC Source when using DIB circuitry to convert the instrument's differential output to a single-ended signal to the DUT pin.

[Single-Ended DUT Pin Only](#)

The figure shows that when using the differential to single-ended DIB circuitry, you can make the required connection by specifying only the device pin name in both the pin map and channel map. This technique is possible because the BBAC Source Positive, Negative, and Reference pins can be programmed using the name of the positive pin. Note however, that the BBAC Source PPMUs that are not identified in the pin map and channel map ([bbacsrcneg](#) and [bbacsrcref](#)) will not be available for use because they are not referenced in the job.



Single-Ended DUT Pin Only

Pin Map for Single-Ended Conversion Connection

Group Name	Pin Name	Type
	A_IN	Analog

Channel Map for Single-Ended Conversion Connection

Pin Name	Package Pin	Type	Site 0	Site 1
A_IN		BBACSrcPos	22.srcpos1	22.srcpos2

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BBAC Source Theory of Operation : Connectivity Options : Single-Ended Device Input Connections

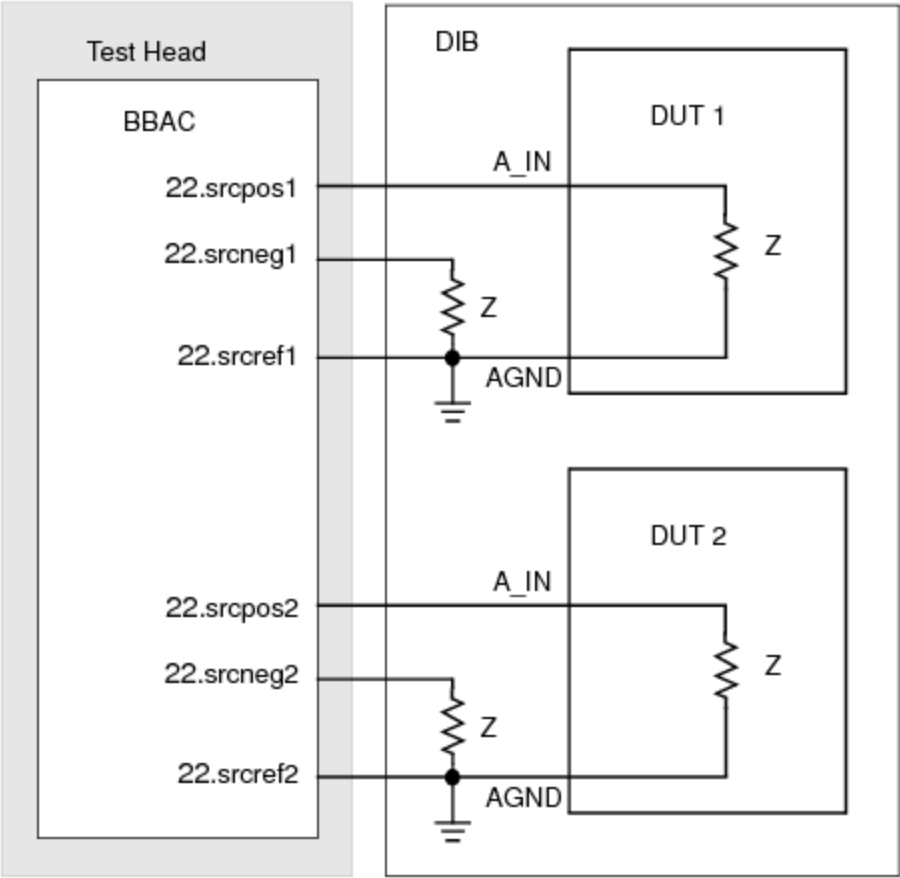
Single-Ended Device Input Connections

Note:

BBAC specifications do not apply when a BBAC Source instrument is used as a single-ended waveform generator.

The figure shows that you can make a single-ended connection to a device:

- | | |
|----|--|
| 1. | Connect the Source Positive pin (srcpos1) to the device input (A_IN). |
| 2. | Connect the Source Reference pin (srcref1) to either the DIB ground or a device reference. |
| 3. | Connect the Source Negative pin (srcneg1) to a load that matches the device input. |



Single-Ended Device Input Connection

The tables show the appropriate pin and channel maps for this configuration. Note that another valid variation of this configuration is to exchange the Source Positive pin with the Source Negative pin. Also note that the figure assumes that the Source Reference pin is connected to the DIB ground.

Pin Map for Single-Ended Connection

Group Name	Pin Name	Type
	A_IN	Analog

Channel Map for Single-Ended Connection

Pin Name	Package Pin	Type	Site 0	Site 1
A_IN		BBACSrcPos	22.srcpos1	22.srcpos2

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BBAC Source Theory of Operation : Connectivity Options : Shared Resources

Shared Resources

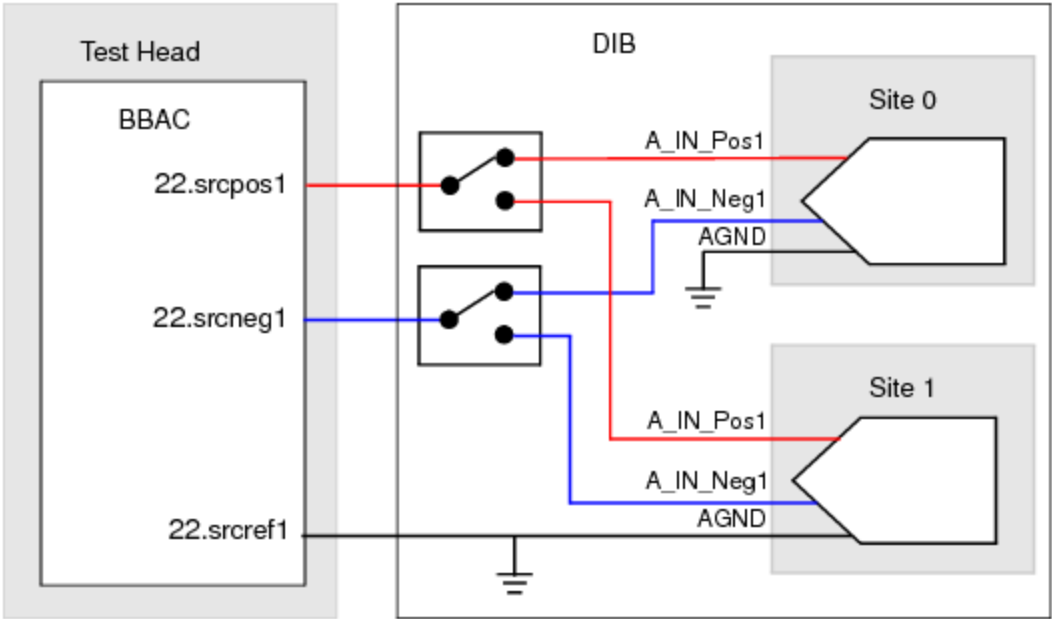
UltraFLEX allows a single instrument to be shared by more than one DUT pin or more than one site in a multisite test environment. However, sharing resources requires that the same instrument channel or pin be used more than once in the channel map. To avoid conflicts, you must modify the standard channel map entries to indicate your intentions.

The following topics describe how resources can be shared across the pins of the same device and across devices in different sites:

- Resources Shared Across Multiple Sites
- Shared Resources Within a Site

Resources Shared Across Multiple Sites

Two or more sites can share a resource by indicating that the same tester channel is connected to more than one site. The first of the sharing sites is the shared reference site. Its tester channel is indicated in the standard way. All subsequent sharing sites use the keyword `site<n>` (for example, `site0`) to indicate the shared reference site with which the resource is being shared.



Resources Shared Across Multiple Sites

Pin Map for Resources Shared Across Multiple Sites

Group Name	Pin Name	Type
	A_IN_Pos1	Analog
	A_IN_Neg1	Analog

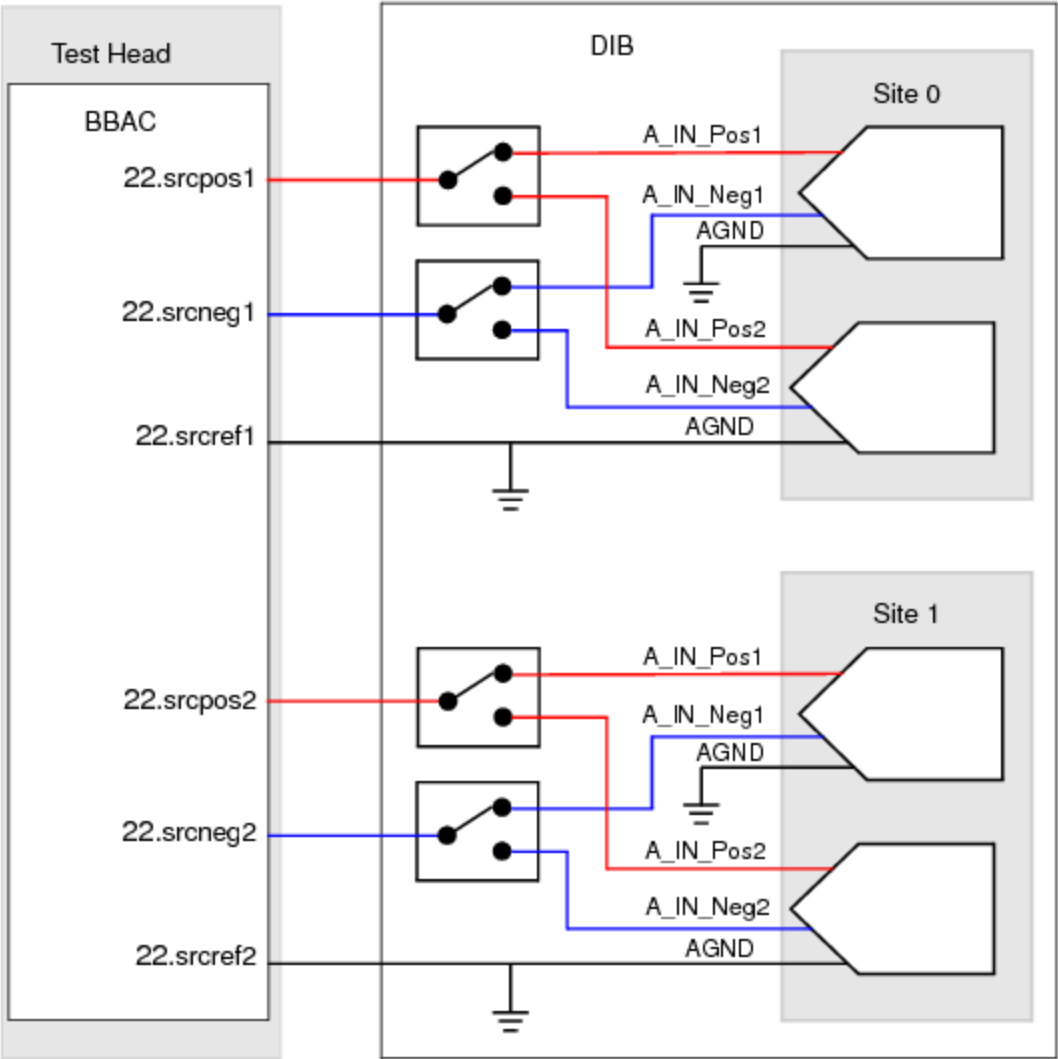
Channel Map for Resources Shared Across Multiple Sites

Pin Name	Package Pin	Type	Site 0	Site 1
A_IN_Pos_1	BBACSrcPos		22.srcpos1	site0
A_IN_Neg_1	BBACSrcNeg		22.srcneg1	site0

Shared Resources Within a Site

A pin can share a resource with other pins by indicating that the same tester channel is connected to both pins. The first pin encountered in the channel map that shares a tester resource designates the tester channel in the standard way. All subsequent pins that share this resource then use the following syntax format to specify their tester channel:

S:<shared-reference-pin>



Shared Resources Within a Site

Pin Map for Shared Resources Within a Site

Group Name	Pin Name	Type
	A_IN_Pos1	Analog
	A_IN_Neg1	Analog
	A_IN_Pos2	Analog
	A_IN_Neg2	Analog

Channel Map for Shared Resources Within a Site

Pin Name	Package Pin	Type	Site 0	Site 1
A_IN_Pos1	BBACSrcPos		22.srcpos1	22.srcpos2

A_IN_Neg1	BBACSrcNeg	22.srcneg1	22.srcneg2
A_IN_Pos2	BBACSrcPos	s:A_IN_Pos1	
A_IN_Neg2	BBACSrcNeg	s:A_IN_Neg1	

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BBAC Source Theory of Operation : Connectivity Options : DIB Access

DIB Access

A test often requires several instrument types to be connected to a DUT pin in a test program. IG-XL supports this requirement in two ways:

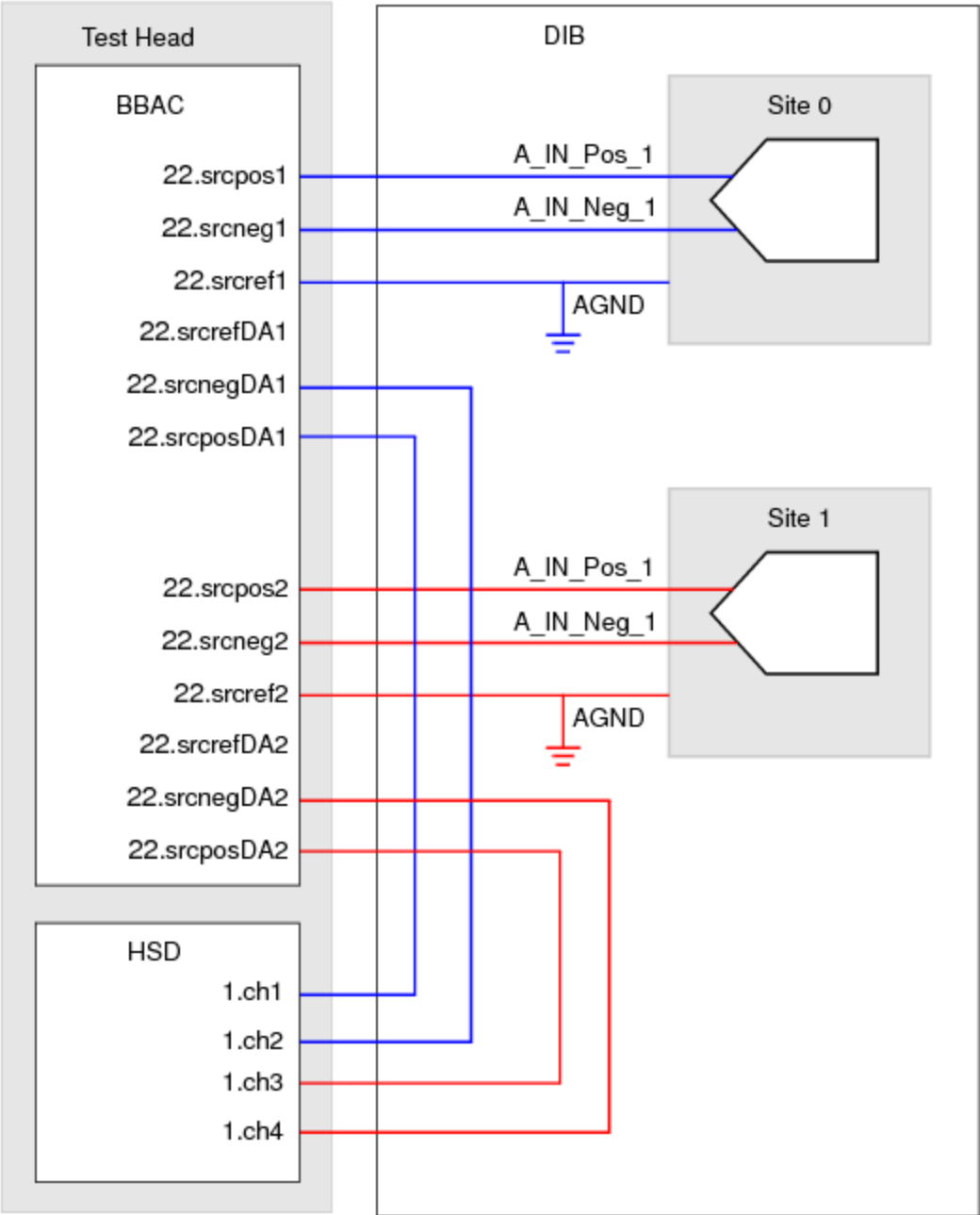
- Directly connecting multiple resources to the DUT pin. See Multiple Resources.
- Using the DIB access feature to connect the DUT pin to one instrument board through another instrument board.

When a device pin needs multiple or different resources, you can use DIB access to minimize the number of DIB relays that are required and to facilitate a simpler programming model. Once the channel map has been set up to indicate the DIB connectivity, IG-XL addresses the correct resource based solely on the pin name. You need not create a virtual pin to reference the different resources.

- BBAC as a Primary Resource
- BBAC Source as Secondary Resource

BBAC as a Primary Resource

When the BBAC Source is used as the primary resource for a device pin, its corresponding DIB Access pins are connected (using DIB traces) to another instrument’s resources. The figure shows HSD resources that are delivered as secondary resources to the device pin using DIB Access.



DIB Access Connection, BBAC Source as a Primary Resource

Pin Map for BBAC Source as a Primary Resource

Group Name	Pin Name	Type
	A_IN_Pos1	Analog
	A_IN_Neg1	Analog

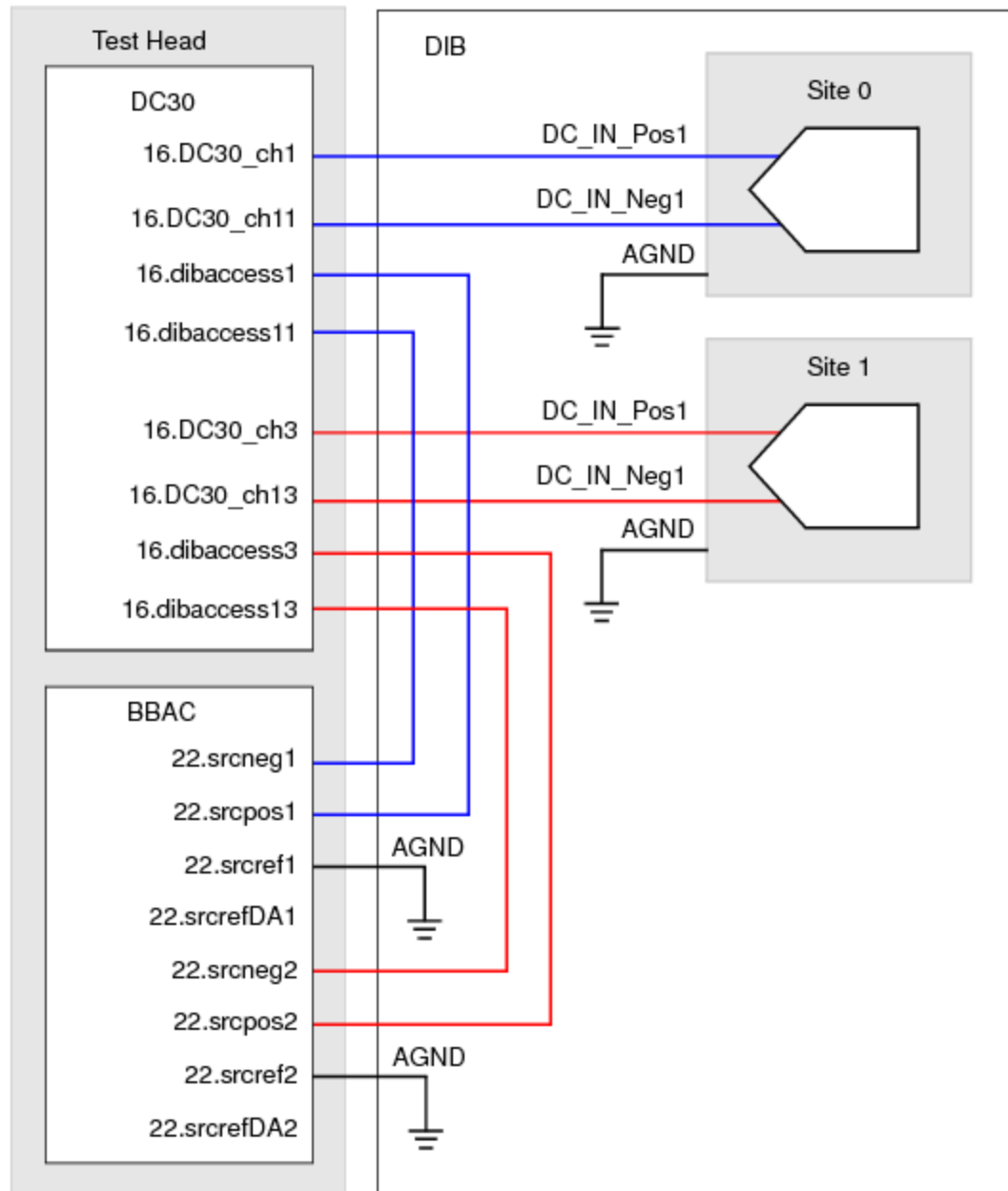
Channel Map for BBAC Source as a Primary Resource

Pin Name	Package Pin	Type	Site 0	Site 1
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A_IN_Pos1	BBACSrcPos	22.srcpos1	22.srcpos2
A_IN_Pos1:DA	I/O	1.ch1	1.ch3
A_IN_Neg1	BBACSrcNeg	22.srcneg1	22.srcneg2
A_IN_Neg1:DA	I/O	1.ch2	1.ch4

BBAC Source as Secondary Resource

The DIB Access use model for the BBAC Source as a secondary resource for a device pin is identical to when it is a primary resource. The only differences are the DIB connectivity and the channel map setup. The figure shows a BBAC Source delivered as a secondary resource to the device pins using DIB Access. Note that the figure may not accurately reflect the DIB Access capability of the DC30.



BBAC DIB Access, BBAC Source as a Secondary Resource

Pin Map for BBAC Source as a Secondary Resource

Group Name	Pin Name	Type
	DC_IN_Pos1	Analog
	DC_IN_Neg1	Analog

Channel Map for BBAC Source as a Secondary Resource

Pin Name	Package Pin	Type	Site 0	Site 1
DC_IN_Pos1		DCVI	16.sense1	16.sense3
DC_IN_Pos1:DA		BBACSrcPos	22.srcpos1	22.srcpos2
DC_IN_Neg1		DCVI	16.sense11	16.sense13
DC_IN_Neg1:DA		BBACSrcNeg	22.srcneg1	22.srcneg2

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BBAC Source Theory of Operation : Connectivity Options : Multiple Resources

Multiple Resources

A test often requires several instrument types to be connected to a DUT pin in a test program. IG-XL supports this requirement in two ways:

- [Directly connecting multiple resources to the DUT pin](#)
- [Using the DIB access feature to connect the DUT pin to one instrument board through another instrument board. See \[DIB Access\]\(#\).](#)

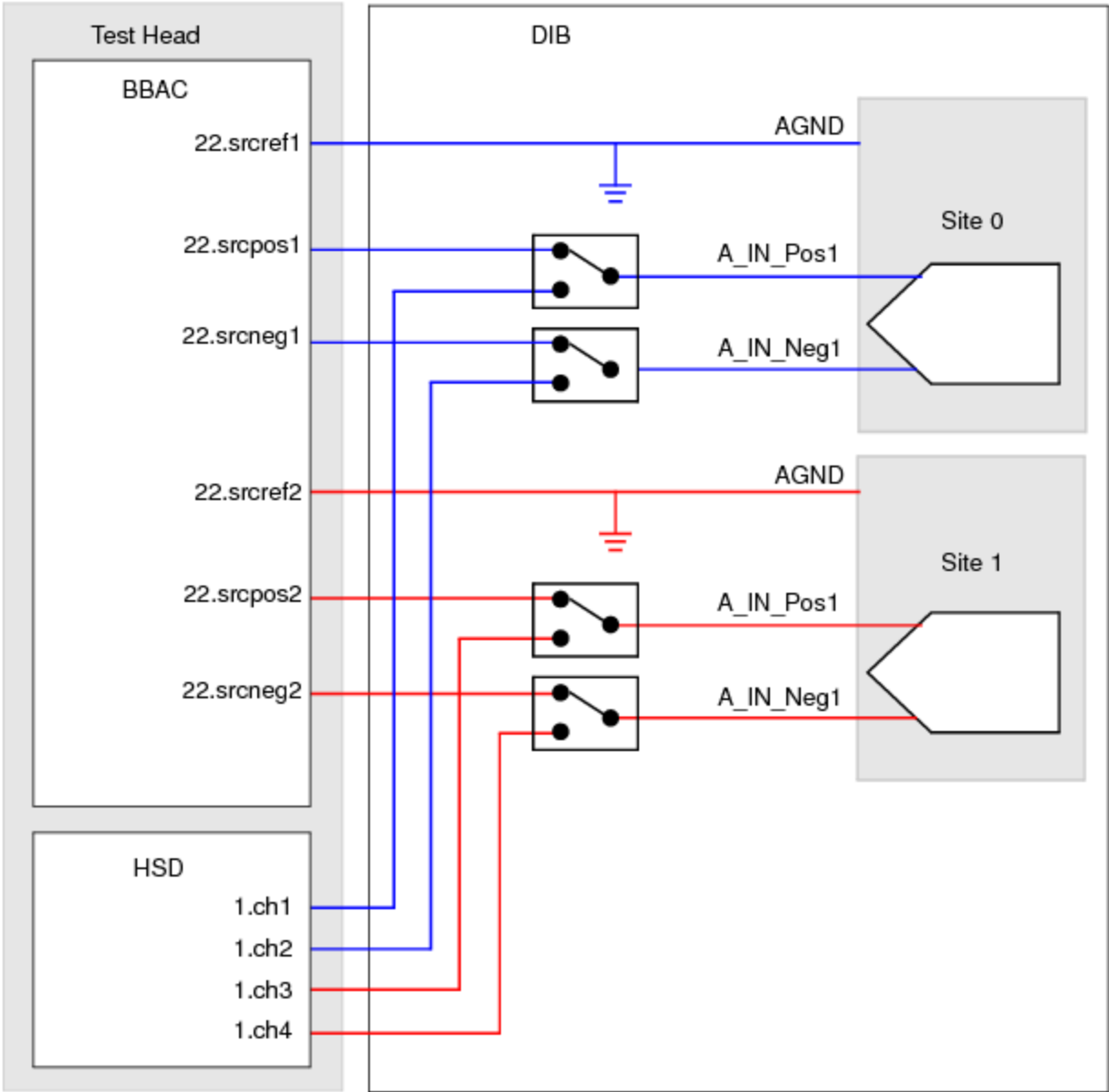
Multiple resources are similar to DIB access with the primary difference being the implied DIB connectivity. Like DIB access, once the channel map has been set up to indicate how the instrument resources have been assigned, IG-XL automatically addresses the correct resource based solely on the pin name. You do not have to create a “virtual” pin to reference the different resources. In most cases of multiple resources, you provide DIB relays to manage instrument connectivity.

[Note from BBAC Source as Primary Resource](#) and [BBAC Source as Secondary Resource](#) that the DIB connectivity is often the same for a BBAC Source as a primary and secondary resource. The significance of the order of the resources in the channel map is when both resources have similar instrumentation (and more importantly language). For example, in the case shown in [BBAC Source as Primary Resource](#), while both the BBAC pins and the HSD pins have PPMU resources available, because the BBAC is listed as the primary resource, its PPMUs are used.

- [BBAC Source as Primary Resource](#)
- [BBAC Source as Secondary Resource](#)

BBAC Source as Primary Resource

The figure shows BBAC Source and HSD channels being connected to the same device input pins. In the corresponding pin map and channel map, since the BBAC Source is listed first, it is the primary resource. Accordingly, if the PPMU is programmed for pins A_IN_Pos1 and A_IN_Neg1, the BBAC's PPMUs are used. Note that the BBAC Source's Type in the Pin Map is defined as Analog.



BBAC Source as Primary Resource

Pin Map for BBAC Source as Primary Resource

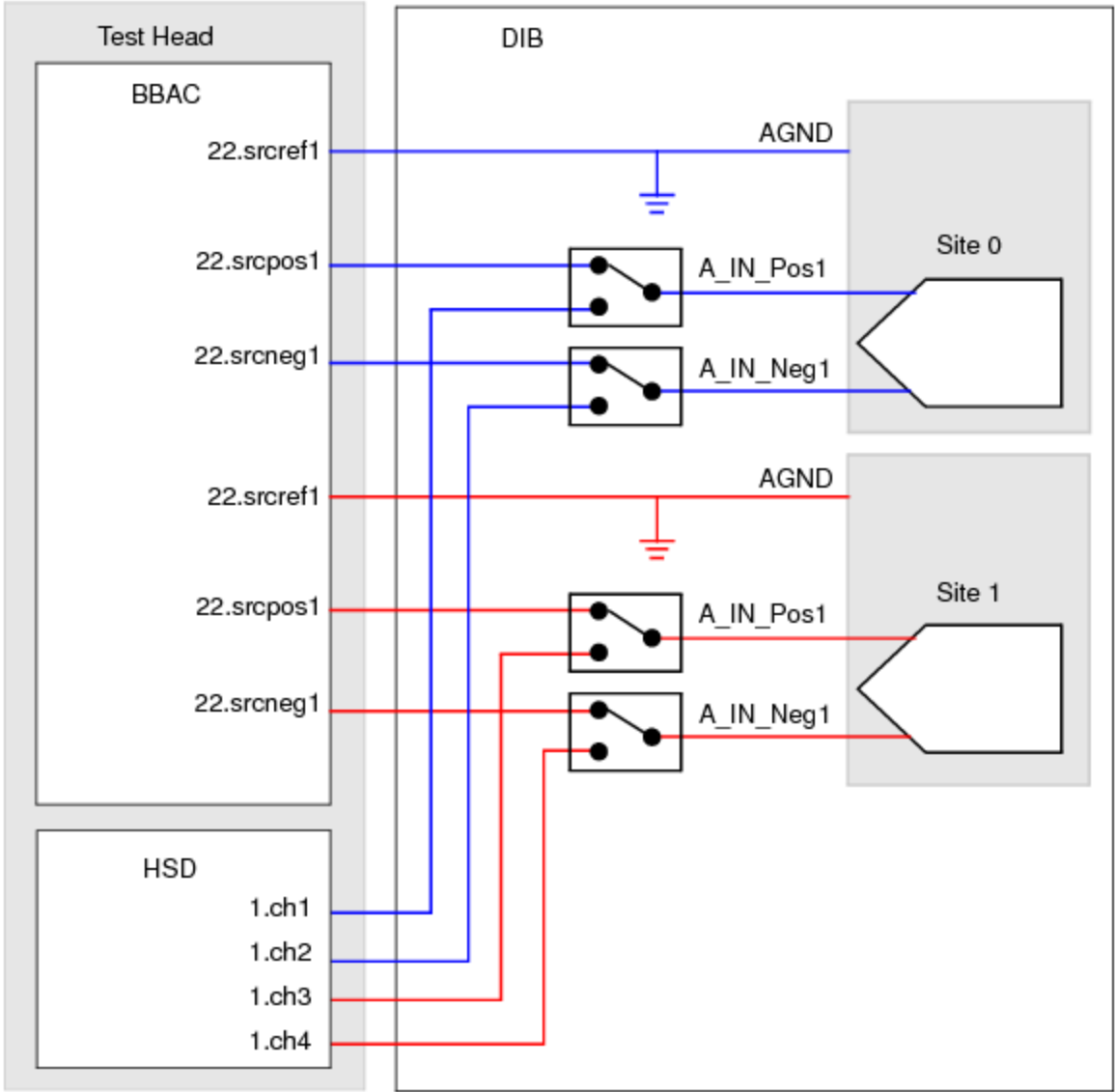
Group Name	Pin Name	Type
	A_IN_Pos1	Analog
	A_IN_Neg1	Analog

Channel Map for BBAC Source as Primary Resource

Pin Name	Package Pin	Type	Site 0	Site 1
A_IN_Pos1		BBACSrcPos	22.srcpos1	22.srcpos2
A_IN_Pos1:M		I/O	1.ch1	1.ch3
A_IN_Neg1		BBACSrcNeg	22.srcneg1	22.srcneg2
A_IN_Neg1:M		I/O	1.ch2	1.ch4

BBAC Source as Secondary Resource

The figure shows BBAC Source and HSD channels being connected to the same device input pins. Note that in this case, because the primary resource for the device pin is an HSD channel, in the pin map the Type is defined as I/O.



BBAC Source as Secondary Resource

Pin Map for BBAC Source as Secondary Resource

Group Name	Pin Name	Type
	A_IN_Pos1	I/O
	A_IN_Neg1	I/O

Channel Map for BBAC Source as Secondary Resource

Pin Name	Package Pin	Type	Site 0	Site 1
A_IN_Pos1		I/O	1.ch1	1.ch3
A_IN_Pos1:M		BBACSrcPos	22.srcpos1	22.srcpos2
A_IN_Neg1		I/O	1.ch2	1.ch4
A_IN_Neg1:M		BBACSrcNeg	22.srcneg1	22.srcneg2

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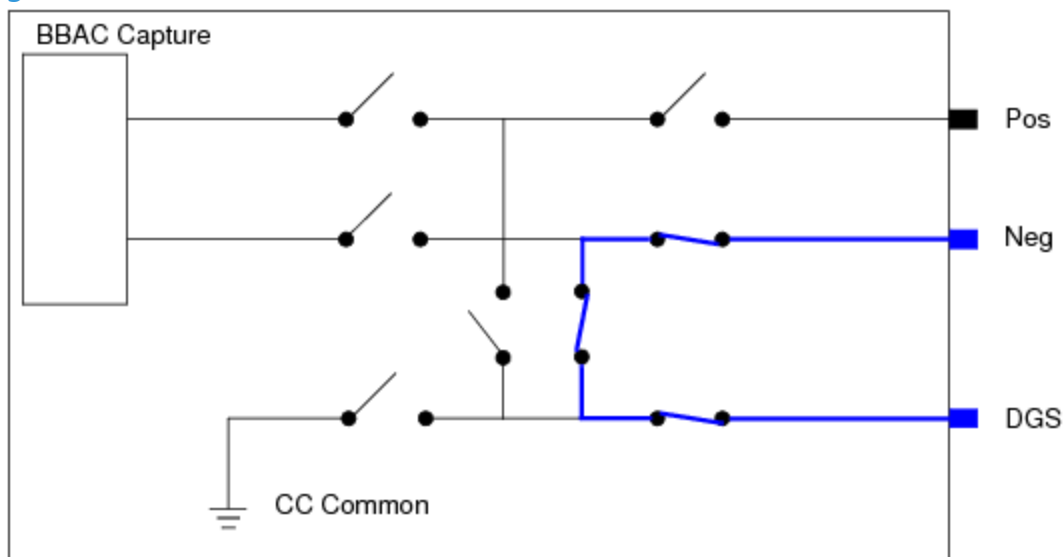
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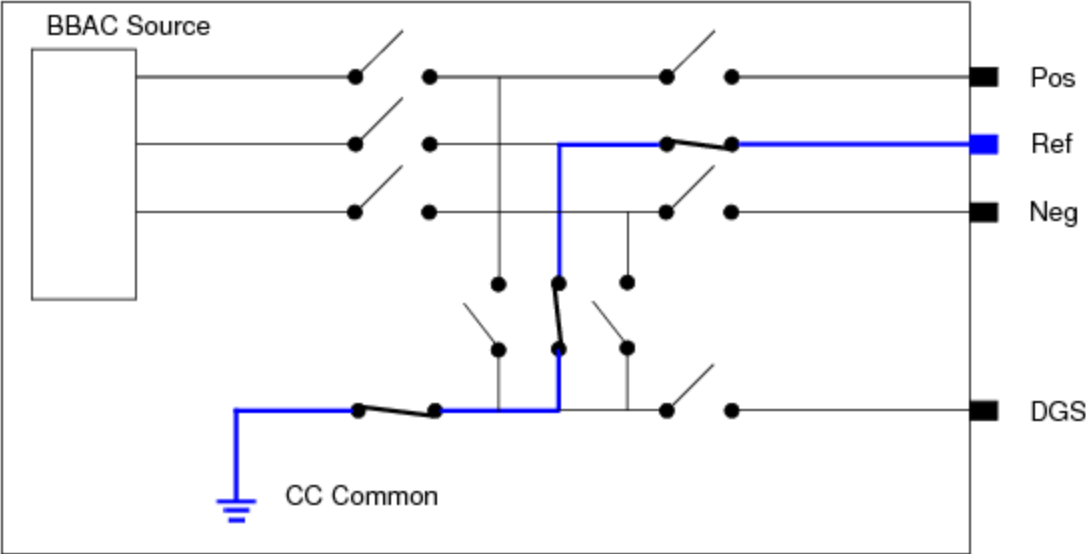
[BBAC Source Theory of Operation : Connectivity Options](#) : Using RefGround to Connect a DUT Pin to Instrument Ground

Using RefGround to Connect a DUT Pin to Instrument Ground

RefGround is a logical instrument that allows you to ground DUT pins by connecting them to a BBAC Source or Capture Channel Card Common (CCC) or Device Ground Sense (DGS). The following figures show how the RefGround instrument uses the BBAC hardware to connect to ground resources.



[RefGround Connects DUT Pin to BBAC Capture DGS Resource](#)



RefGround Connects DUT Pin to BBAC Source CC Common Resource

The RefGround instrument has its own VBT syntax to connect or disconnect DUT pins to or from the BBAC hardware. Use the following syntax to connect to and disconnect from the instrument’s reference ground resources:

TheHdw.RefGround.Pins().Connect(tlCCGroundType)

TheHdw.RefGround.Pins().Disconnect(tlCCGroundType)

where tlCCGroundType can be one of the following optional arguments:

tlCCGroundTypeDefault	The instrument’s default ground resource which is Channel Card Common (CCC).
tlCCGroundTypeCCC	The BBAC’s Channel Card Common ground resource.
tlCCGroundTypeDGS	The BBAC’s Device Ground Sense resource.

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BBAC Source Theory of Operation : User View of DIB Pad Patterns

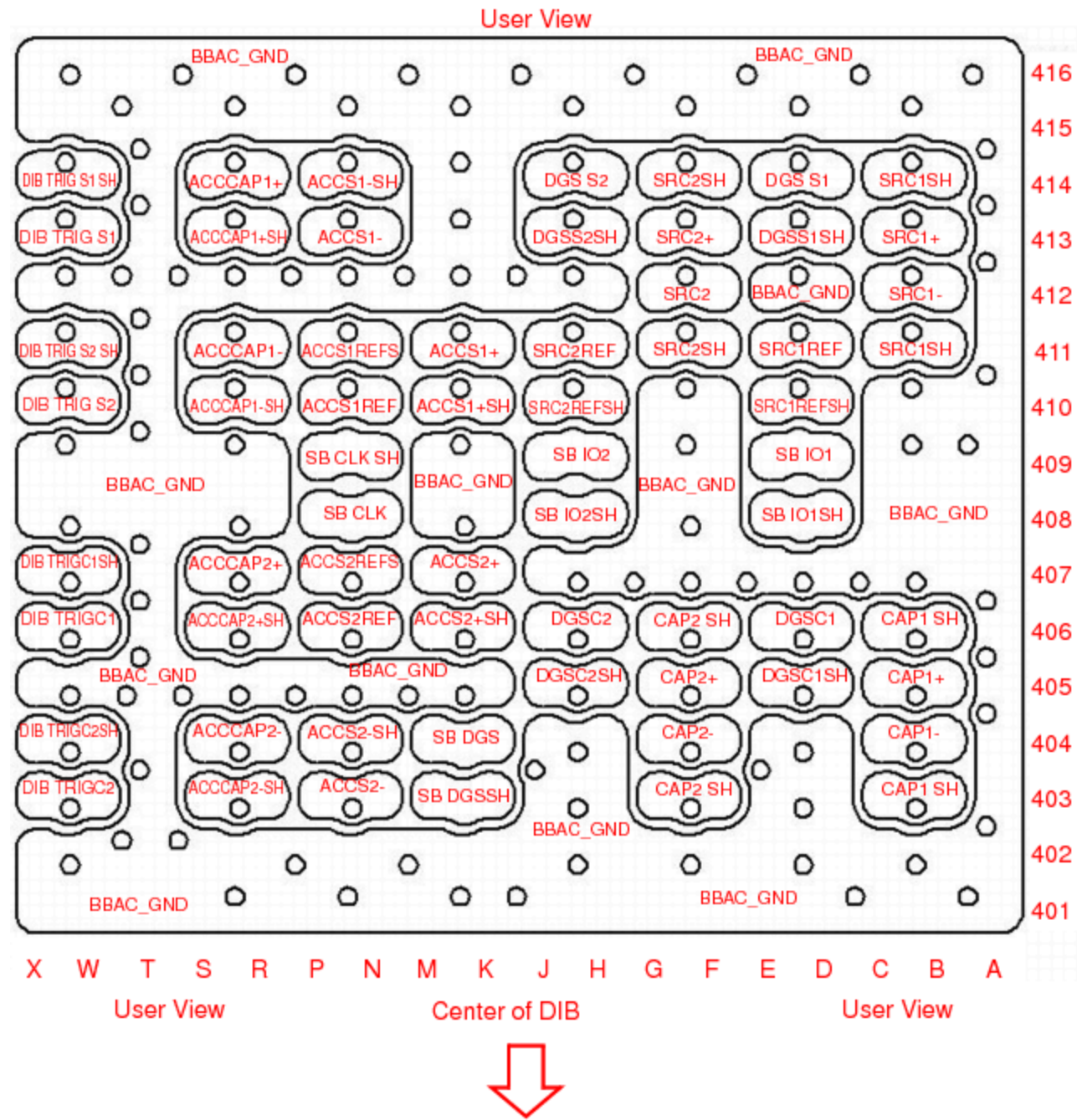
User View of DIB Pad Patterns

This section shows the BBAC pad patterns as seen from the user perspective, looking at the topside (device) of the DIB/HIB/PIB and towards the test system.

☑ Note:

When designing a DIB, use the tester view of the pad patterns (looking at the stiffener side of the DIB/HIB/PIB), which is given in the [BBAC Signal Pad Pattern \(Tester View\)](#) because all the artwork and design kits use that view.

The following figure shows the BBAC pad pattern located on sidewalk 4.



BBAC Pad Pattern for Sidewalk 4 (User View)